

**ALVA'S INSTITUTE OF ENGINEERING AND TECHNOLOGY
(AIET)**

Department of AIML & ISE

A PROJECT REPORT ON

**AI-POWERED CRIMINAL NETWORK
ANALYSIS SYSTEM**

BACHELOR OF ENGINEERING

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

AND

INFORMATION SCIENCE AND ENGINEERING

Submitted by

Akshay H (USN: 4AL23AI006)

Omkar K S (USN: 4AL23ISE036)

SIH ID: SIH26189

Team: Trailblazers (Team ID: TALVA105)

Under the guidance of

Anjit Raja R

Academic Year: [2026 - 2027]

CERTIFICATE

This is to certify that the project work entitled "**AI-Powered Criminal Network Analysis System**" is a bonafide work carried out by **Akshay H (USN: 4AL23AI006)** and **Omkar K S (USN: 4AL23ISE036)** (Team Trailblazers, Team ID: TALVA105; SIH ID: SIH26189) in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Artificial Intelligence and Machine Learning of Visvesvaraya Technological University, Belagavi, at the Department of AIML & ISE, Alva's Institute of Engineering and Technology, during the academic year 2026 - 2027

It is certified that all corrections and suggestions indicated during internal assessment have been incorporated in the report. The project report satisfies the academic requirements in respect of project work prescribed for the said degree.

Anjit Raja R, Project Guide

DECLARATION

We hereby declare that the project work entitled "**AI-Powered Criminal Network Analysis System**" submitted to the Department of Artificial Intelligence and Machine Learning, Alva's Institute of Engineering and Technology, in partial fulfilment of the requirements for the degree of Bachelor of Engineering, is a record of original work done by us under the guidance of Anjit Raja R.

We further declare that the work reported in this report has not been submitted, in part or in full, for the award of any other degree or diploma. Sources of information, including software documentation and published literature, have been acknowledged in the references.

Place: MIJAR

Date: 30-09-2026

Akshay H (4AL23AI006)

Omkar K S (USN: 4AL23ISE036)

ACKNOWLEDGEMENT

We express our sincere gratitude to our project guide, Anjit Raja R, for guidance and encouragement throughout this work. We thank the Head of the Department of Artificial Intelligence and Machine Learning, [HOD NAME — TO BE PROVIDED], and the Principal of Alva's Institute of Engineering and Technology, [PRINCIPAL NAME — TO BE PROVIDED], for providing the academic environment and facilities for this project.

We are grateful to the faculty and staff of the department, and to our friends and families for their support. We also acknowledge the developers and maintainers of the open-source tools and platforms used in this project, including React, Vite, Flask, PostgreSQL, Gunicorn, Groq and Vercel.

Akshay H

Omkar K S

ABSTRACT

Crime records held by law-enforcement agencies contain implicit relationships among cases, accused persons, victims, officers, stations and districts, but these relationships are difficult to examine in record-by-record views. This project presents the AI-Powered Criminal Network Analysis System, a web-based crime intelligence platform that centralises such records and provides analytical, spatial and relationship views.

The system comprises a React and TypeScript frontend built with Vite, a Flask REST backend served with Gunicorn, a PostgreSQL production database accessed through psycopg2 with connection pooling, and an AI layer that calls the Groq API (default model openai/gpt-oss-120b, with configured fallback models). It is deployed on Vercel, with the Flask application exposed as a Python serverless function. The backend offers endpoints for cases, key performance indicators, trends, districts, stations, crime categories, demographics, officers, accused persons, victims, AI predictions, hotspots, forecasts, district risk, anomalies, similar cases, the criminal network, AI status and AI chat.

A criminal linkage graph presents accused persons, victims, cases, crime heads, stations and courts as typed nodes and edges, allowing repeated involvement and inter-case connections to be inspected visually. The AI layer is designed as decision support: its output requires human verification and does not establish guilt.

The production endpoint GET /api/cases/kpis was tested and returned HTTP 200 with 120 FIRs, 110 solved, 59 closed, 10 pending, 51 charge-sheeted, 79 accused, 120 victims, 7 districts, 29 police stations and 1 FIR registered on the day of testing. These are database statistics and not machine-learning accuracy values. No model accuracy evaluation was performed. Limitations include a small dataset, dependence on data quality, the need for rigorous authorisation, and the absence of formal AI evaluation. Future work includes graph algorithms, explainable AI, role-based access control and audit logging.

TABLE OF CONTENTS

CHAPTER 1 – INTRODUCTION	1
1.1 Background	1
1.2 Motivation.....	2
1.3 Problem Statement	2
1.4 Need for the System.....	2
1.5 Objectives	3
1.6 Scope.....	3
1.7 Contributions.....	4
1.8 Organization of the Report.....	4
CHAPTER 2 – LITERATURE SURVEY	5
2.1 Introduction.....	5
2.2 Crime Data Analytics.....	5
2.3 Artificial Intelligence in Crime Analysis	5
2.4 Criminal Network Analysis	5
2.5 Crime Prediction and Hotspot Analysis.....	6
2.6 AI-assisted Investigation Systems	6
2.7 Existing Approaches	6
2.8 Limitations of Existing Approaches	7
2.9 Research/Project Gap.....	7
2.10 Summary	7
CHAPTER 3 – SYSTEM ANALYSIS.....	8
3.1 Existing System	8
3.2 Problems with Existing System	8
3.3 Proposed System.....	8
3.4 Advantages of Proposed System.....	9
3.5 Functional Requirements	9
3.6 Non-Functional Requirements	10
3.7 Hardware Requirements.....	10
3.8 Software Requirements.....	10
3.9 Feasibility Study	11
CHAPTER 4 – SYSTEM DESIGN	12
4.1 System Architecture.....	12
4.2 High-Level Architecture	12
4.3 Frontend Architecture	13
4.4 Backend Architecture.....	13

4.5 Database Architecture	13
4.6 AI Service Architecture	14
4.7 API Architecture	15
4.8 Data Flow.....	15
4.9 Module Design.....	16
4.10 Security Considerations	17
4.11 System Workflows.....	17
CHAPTER 5 – DATABASE DESIGN	19
5.1 Database Introduction	19
5.2 Database Selection	19
5.3 Entity Relationship Concept	19
5.4 Entity Relationships.....	20
5.5 Database Schema	20
5.6 Important Entities.....	20
5.7 Relationships.....	21
5.8 Data Integrity	21
5.9 Connection Pooling.....	21
5.10 Database Security.....	22
CHAPTER 6 – IMPLEMENTATION.....	23
6.1 Implementation Overview	23
6.2 Frontend Implementation.....	23
6.3 Backend Implementation	24
6.4 REST API Implementation	24
6.5 Database Integration	24
6.6 AI Integration.....	24
6.7 CORS Configuration.....	25
6.8 Error Handling	25
6.9 API Request/Response Flow.....	25
6.10 Deployment Implementation	25
CHAPTER 7 – MODULES	27
7.1 Dashboard (Command Center)	27
7.2 Case Analytics (FIR Management).....	28
7.3 Case Details	29
7.4 KPI Analytics.....	29
7.5 District Analysis.....	30
7.6 Police Station Analysis	30
7.7 Accused Analysis.....	30

7.8 Victim Analysis	31
7.9 Officer Analysis	32
7.10 Crime Categories	33
7.11 Crime Trends	33
7.12 AI Predictions	34
7.13 Hotspot Analysis.....	34
7.14 Anomaly Analysis.....	35
7.15 Similar Case Analysis	36
7.16 Criminal Network Analysis	36
7.17 AI Chat.....	37
7.18 Supporting Module: User Access and Approval	38
CHAPTER 8 – AI AND ANALYTICS	40
8.1 AI Architecture	40
8.2 Groq API Integration	40
8.3 AI Chat.....	40
8.4 Crime Trend Analysis	40
8.5 Hotspot Analysis.....	40
8.6 Anomaly Analysis.....	41
8.7 Similar Case Analysis	41
8.8 District Risk Analysis	41
8.9 Criminal Network Analysis	41
8.10 Limitations of AI-based Analysis	42
8.11 Ethical Considerations	42
CHAPTER 9 – TESTING.....	44
9.1 Testing Introduction.....	44
9.2 Testing Strategy	44
9.3 Unit Testing	44
9.4 Integration Testing	44
9.5 API Testing	44
9.6 Database Testing.....	45
9.7 Frontend Testing	45
9.8 CORS Testing	45
9.9 Deployment Testing.....	45
9.10 Test Cases	45
CHAPTER 10 – RESULTS AND DISCUSSION.....	47
10.1 System Results.....	47
10.2 Production API Results.....	47

10.3 Dashboard Results	47
10.4 Database Results	48
10.5 AI Functionality	48
10.6 Deployment Results	48
10.7 Discussion	49
CHAPTER 11 – ADVANTAGES, LIMITATIONS AND APPLICATIONS	50
11.1 Advantages.....	50
11.2 Limitations	50
11.3 Applications	50
11.4 Ethical and Legal Considerations	51
CHAPTER 12 – FUTURE ENHANCEMENT	52
CHAPTER 13 – CONCLUSION	53
REFERENCES	54
APPENDICES	Error! Bookmark not defined.
Appendix A – Important API Endpoints	Error! Bookmark not defined.
Appendix B – Important Code Snippets	Error! Bookmark not defined.
Appendix C – Database Information	Error! Bookmark not defined.
Appendix D – Testing Evidence	Error! Bookmark not defined.
Appendix E – Screenshots	Error! Bookmark not defined.
SUPPLEMENTARY SECTIONS	Error! Bookmark not defined.
A. Project Synopsis.....	Error! Bookmark not defined.
B. Abstract (250–300 words).....	Error! Bookmark not defined.
C. Key Contributions	Error! Bookmark not defined.
D. Viva Preparation – 28 Likely Questions	Error! Bookmark not defined.
E. Five-Minute Presentation Speech.....	Error! Bookmark not defined.
F. Ten-Minute Presentation Speech.....	Error! Bookmark not defined.
G. Glossary	Error! Bookmark not defined.

LIST OF FIGURES

- Figure 1: Overall System Architecture
- Figure 2: Frontend-Backend Architecture
- Figure 3: Database Architecture
- Figure 4: AI Integration Architecture
- Figure 5: API Request Flow (GET /api/cases/kpis)
- Figure 6: Sign-in Page
- Figure 7: Account Request Form with Requested Role Selection
- Figure 8: Production Deployment: Backend Project on Vercel
- Figure 9: Production Deployment: Frontend Project on Vercel
- Figure 10: Dashboard (Command Center)
- Figure 11: FIR Records Management with Crime Records Menu
- Figure 12: KPI Analytics: Dashboard KPI Cards and Monthly Crime Volume
- Figure 13: Accused and Repeat Offender Analytics
- Figure 14: Victim Analytics
- Figure 15: Officer Performance Registry
- Figure 16: Crime Analytics: Monthly Trend, District, Sub-head and Victim Gender Charts
- Figure 17: AI Intelligence Modules: Crime Hotspot Prediction
- Figure 18: Interactive Spatial Heatmap of Crime Incidents
- Figure 19: Criminal Linkage Network: Isolated Suspect Network
- Figure 20: Criminal Linkage Network: Suspect with Multiple Linked Cases
- Figure 21: AI Intelligence Assistant (Chat)
- Figure 22: User Approval Center
- Figure 23: Criminal Network Relationship Diagram (Conceptual)

LIST OF TABLES

Table 1: Technology Stack

Table 2: Summary of Reviewed Approaches

Table 3: Functional Requirements

Table 4: Non-Functional Requirements

Table 5: API Categories

Table 6: System Modules

Table 7: Database Entities (Conceptual)

Table 8: Test Cases

Table 9: Production KPI Results (Verified System/Database Statistics)

Table 10: Implemented, Verified and Future Functionality

Table 11: Advantages

Table 12: Limitations

Table 13: Future Enhancements (Not Implemented)

Table 14: Endpoint Inventory

Table 15: Snippets to Insert

Table 16: Glossary of Terms

LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
AIET	Alva's Institute of Engineering and Technology
AIML	Artificial Intelligence and Machine Learning
API	Application Programming Interface
CORS	Cross-Origin Resource Sharing
FIR	First Information Report
GNN	Graph Neural Network
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
KPI	Key Performance Indicator
KSP	Karnataka Police (as identified by the application's own dashboard subtitle)
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
PS	Police Station
RBAC	Role-Based Access Control
REST	Representational State Transfer
SIH	Smart India Hackathon
SQL	Structured Query Language
TLS	Transport Layer Security
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator
USN	University Seat Number
WSGI	Web Server Gateway Interface

CHAPTER 1 – INTRODUCTION

1.1 Background

Law-enforcement agencies generate large volumes of structured records in the course of registering and investigating crime. A First Information Report (FIR) links a registered case to a police station, a district, a crime category, an investigating officer, one or more accused persons and one or more victims. Individually, each record describes a single incident; collectively, the records form a network of people, places and events in which repeated involvement, shared locations and recurring crime types may be significant. Research on crime data mining has long argued that entity relationships, and not only aggregate counts, are a valuable source of investigative insight [1], [2].

Network analysis provides a formal vocabulary for such relationships. Sparrow discussed how concepts from network analysis, such as centrality and structural position, can be applied to criminal intelligence [3], while standard texts describe the mathematical basis of graph-based analysis [4], [5]. In parallel, recent large language models (LLMs) based on the Transformer architecture [6] allow natural-language summaries and question answering over structured data, provided that their outputs are treated with appropriate caution [7].

This project, the **AI-Powered Criminal Network Analysis System**, is a web-based crime intelligence platform that combines a React and TypeScript frontend, a Flask REST backend, a PostgreSQL database and an AI layer built on the Groq API, deployed on Vercel. It is an academic prototype developed for the B.E. degree in Artificial Intelligence and Machine Learning at the Department of AIML & ISE, Alva's Institute of Engineering and Technology, and carries the SIH ID SIH26189. It was developed by Team Trailblazers (Team ID: TALVA105). In the application the platform is presented as the KSP AI-Portal, with the dashboard subtitle "Karnataka Police AI Crime Intelligence Dashboard", which an authorised user reaches after selecting a security role (Section 3.3).

Status labels used in this report

To keep the report faithful to the actual system, three labels are used throughout: **Implemented** (the functionality exists in the delivered system, supported by the project description or by screenshots supplied for this report); **Verified** (the functionality was tested and a result was recorded); and **Future** (proposed work that is not part of the current

system). Information that could not be established from the project material is written as [TO BE PROVIDED].

1.2 Motivation

Crime records are often accessed through record-level screens or spreadsheets, which makes it laborious to answer analytical questions such as which districts show rising activity, whether an accused person appears in several cases, or which cases resemble one another. The motivation for this project is to provide a single web interface in which such questions can be explored through dashboards, maps, relationship graphs and an AI-assisted query facility, while keeping the underlying data in a relational database that supports integrity and concurrent access.

1.3 Problem Statement

To design and implement a web-based platform that centralises crime-related records (cases, accused persons, victims, officers, districts, police stations and crime categories), exposes them through a REST API, presents analytical views (key performance indicators, trends, hotspots, anomalies, similar cases and criminal relationships) and augments these views with an AI-assisted intelligence layer, such that the outputs serve as decision support for a human analyst and never as a determination of guilt.

1.4 Need for the System

- **Fragmented information:** case, accused, victim and station data are viewed separately, obscuring cross-case relationships.
- **Manual analysis effort:** trend and hotspot questions require repeated manual aggregation.
- **Relationship visibility:** repeated involvement of an accused person in multiple cases is difficult to see in tabular form but natural to see in a graph [2].
- **Accessibility:** a browser-based system with a documented REST interface can be used without installing client software.
- **Manual cross-referencing:** connecting an accused person, a location or a legal section across cases relies on individual memory or manual search, which does not scale as the number of records grows.

- **Lack of interactive visualisation:** paper or tabular records do not lend themselves to exploring a network of connections.

1.5 Objectives

1. To centralise crime-related information in a PostgreSQL database.
2. To provide a web-based crime intelligence dashboard.
3. To enable efficient analysis of crime cases.
4. To analyse accused and victim information.
5. To analyse district-level and police-station-level crime information.
6. To provide AI-assisted intelligence through the Groq API.
7. To identify crime trends and potential hotspots.
8. To support anomaly analysis and similar-case analysis.
9. To visualise criminal relationships as a network.
10. To provide an accessible REST-based backend architecture.

1.6 Scope

The scope covers the implemented web application and its deployment. It includes the React and TypeScript frontend, the Flask backend, PostgreSQL integration, Groq-based AI functions and the deployment on Vercel. It excludes integration with any real police information system, real-time data feeds, formal machine-learning model training and evaluation, and legal or evidentiary use of the outputs. The technologies used are summarised in Table 1.

Table 1: Technology Stack

Layer	Technology	Role in the system
Frontend	React, TypeScript, Vite	User interface, dashboards, visualisations; build tooling by Vite
Backend	Python, Flask, Flask-CORS, Gunicorn	REST API, cross-origin handling, production WSGI serving
Database	PostgreSQL, psycopg2, connection pooling	Production relational store; pooled access from the backend
AI	Groq API	AI-assisted analysis and chat (default model openai/gpt-oss-120b)
Deployment	Vercel	Hosting; Flask exposed as a Python serverless function

Layer	Technology	Role in the system
Other	REST, JSON, CORS, Git/GitHub	Interface style, data format, cross-origin policy, version control

1.7 Contributions

- A full-stack crime analytics platform integrating React/TypeScript, Flask, PostgreSQL and the Groq API.
- A REST API covering cases, KPIs, trends, districts, stations, categories, demographics, officers, accused, victims, hotspots, forecasts, district risk, anomalies, similar cases, criminal network, AI status and AI chat.
- An interactive criminal linkage graph connecting accused, victims, cases, crime heads, stations and courts.
- A production deployment on Vercel with a verified production KPI endpoint.
- A discussion of the ethical limits of AI-assisted crime analysis.

1.8 Organization of the Report

Chapter 2 reviews the literature. Chapter 3 analyses the system and its requirements. Chapter 4 presents the design and Chapter 5 the database design. Chapter 6 describes the implementation and Chapter 7 the modules. Chapter 8 covers the AI and analytics layer. Chapter 9 presents testing and Chapter 10 the results. Chapter 11 discusses advantages, limitations and applications; Chapter 12 outlines future work; Chapter 13 concludes. References and appendices follow.

CHAPTER 2 – LITERATURE SURVEY

2.1 Introduction

This chapter reviews published work relevant to the system: crime data analytics, artificial intelligence in crime analysis, criminal network analysis, hotspot and prediction methods, and AI-assisted investigation. Only sources that can be verified are cited. Descriptions are kept at the level of the sources' well-established contributions.

2.2 Crime Data Analytics

Chen et al. proposed a general framework for crime data mining that covers entity extraction, clustering, association analysis, classification and social network analysis, illustrated through law-enforcement applications [1]. The framework established that crime records benefit from a pipeline that moves from raw records to entities and then to relationships. The present system follows the same broad progression: relational records are retrieved, aggregated into indicators and trends, and then related through a network view.

2.3 Artificial Intelligence in Crime Analysis

The RAND study on predictive policing surveys forecasting methods and emphasises that predictions support, rather than replace, operational judgement [8]. More recent language-model technology rests on the Transformer architecture [6], which enables text generation and question answering. A known limitation of such models is hallucination, the generation of fluent but unsupported content [7]. Retrieval-augmented generation grounds a model's output in retrieved documents and is one recognised mitigation [9]. Bias in predictive policing has also been documented: Lum and Isaac showed that models trained on recorded crime can reproduce patterns of past recording practice [10]. These findings motivate the decision-support framing adopted in this report.

2.4 Criminal Network Analysis

Sparrow assessed the prospects of applying network analysis to criminal intelligence and discussed structural concepts such as centrality and the vulnerability of networks to removal of key actors [3]. Xu and Chen presented a framework for criminal network analysis and visualisation that combines network partitioning, structural measures, role

analysis and visual display, and applied it to law-enforcement data [2]. Standard textbooks give the graph-theoretic and sociological foundations [4], [5]. The present system implements the relationship-and-visualisation aspects (an entity graph with typed edges) but does not implement centrality computation or community detection; these are listed as future work.

2.5 Crime Prediction and Hotspot Analysis

Sherman et al. reported that predatory crime concentrates in a small number of places, which underpins hotspot-oriented analysis [11]. Mohler et al. modelled crime with self-exciting point processes, capturing the tendency of crime events to cluster in space and time [12]. Anomaly detection, surveyed by Chandola et al., identifies point, contextual and collective anomalies in data [13]. These works provide the conceptual background for the hotspot, forecast and anomaly modules. The present system does not claim to implement the cited statistical models; its hotspot and anomaly outputs are produced by its own backend logic and the AI layer, whose exact computations are [TO BE PROVIDED] for the final report.

2.6 AI-assisted Investigation Systems

Systems in the COPLINK tradition integrated records across agencies and offered analytical tools to investigators [1], [2]. Modern equivalents add conversational interfaces. The literature cautions that such interfaces require grounding and human verification [7], [9], and that recorded data may carry bias [10].

2.7 Existing Approaches

Table 2: Summary of Reviewed Approaches

Approach	Source	Contribution
Crime data mining framework	[1]	Entity extraction, clustering, association, classification, network analysis
Criminal network analysis and visualisation	[2]	Network partition, structural measures, role analysis, visualisation
Network analysis for intelligence	[3]	Applicability of centrality and structural concepts
Hot-spot criminology	[11]	Concentration of crime in small places
Self-exciting point processes	[12]	Space-time clustering model of crime events
Anomaly detection survey	[13]	Taxonomy of anomaly types and techniques
Predictive policing overview	[8]	Forecasting methods and operational role

2.8 Limitations of Existing Approaches

- Many analytical tools are desktop or agency-internal and are not documented as browser-based REST services.
- Statistical forecasting models require careful data preparation and evaluation and can reproduce recording bias [10].
- Language models may produce unsupported statements [7], so their outputs cannot be treated as facts.
- Network analysis outputs indicate association, not guilt [3].

2.9 Research/Project Gap

The literature separates data mining, network analysis and language-model assistance. The gap addressed here is an integrated, web-deployed prototype that places these views behind one REST API on a relational database, with an explicit decision-support stance. The project does not claim novelty in algorithms; its contribution is integration and deployment.

2.10 Summary

The reviewed work supports three design choices: relational storage of entities, graph presentation of relationships, and AI assistance under human oversight. It also supplies the caveats (bias, hallucination, association versus guilt) that shape Chapters 8 and 11.

CHAPTER 3 – SYSTEM ANALYSIS

3.1 Existing System

The project material does not describe a specific legacy system. In general, the existing arrangement for analysis of crime records consists of record-level viewing screens, spreadsheets and manual report preparation. The details of any particular agency's current tooling are [TO BE PROVIDED].

3.2 Problems with Existing System

- Case, accused and victim information is examined in separate views.
- Trend, hotspot and comparison questions require manual aggregation.
- Relationships across cases are not visualised.
- No conversational interface for querying the data.
- Connecting an accused person or location across cases relies on manual search or individual memory.
- The volume of records grows continuously, increasing the manual effort of review.
- Producing crime statistics or trend summaries by hand is time-consuming.

These are general limitations of manual and register-based approaches; they are not a claim about any specific named police or government system, none of which was evaluated for this report.

3.3 Proposed System

The proposed system is a three-tier web application. A React and TypeScript frontend communicates through a centralised API service with a Flask REST backend, which reads from PostgreSQL and, for AI-assisted functions, calls the Groq API. The frontend obtains the API base address from the VITE_API_URL environment variable. The system is deployed on Vercel with Flask exposed as a Python serverless function.

Access to the portal is preceded by role selection. The "Switch Security Role" menu in the interface lists five named roles: Administrator, Police Officer, Investigation Officer, Crime Analyst and District Superintendent. The account-request form offers the last four (Section 4.10), and Administrator is the role shown as active in the captured session. A global search box in the top bar covers crime number, accused, officer and court fields. That the roles are

selectable and that the search box is present are interface observations; whether each role enforces a distinct, restricted set of permissions on the server (role-based access control), and how search results are returned, are [TO BE PROVIDED].

3.4 Advantages of Proposed System

- Single browser-based entry point for case analytics, maps, AI views and network graphs.
- Separation of presentation, API and data layers.
- Relational integrity and concurrent access through PostgreSQL [14].
- Documented REST interface usable by other clients [15].
- AI-assisted summaries and chat as a supplement to conventional analytics.

3.5 Functional Requirements

Table 3: Functional Requirements

ID	Requirement	Status
FR-01	Retrieve and list FIR/case records with filters (crime category, status, search)	Implemented (screenshot)
FR-02	Display case details	Implemented (endpoint listed)
FR-03	Compute and display KPIs (FIRs, solved, closed, pending, charge-sheeted, accused, victims, districts, stations, today's FIRs)	Implemented; endpoint Verified
FR-04	Provide trend, district, station, category and demographic analytics	Implemented (screenshot)
FR-05	Provide accused, victim and officer views	Implemented (screenshot/endpoint list)
FR-06	Provide AI predictions, hotspots, forecast, district risk, anomalies and similar cases	Implemented (endpoint list; hotspot screen shown)
FR-07	Provide criminal network graph with case- and suspect-level isolation	Implemented (screenshot)
FR-08	Provide AI status and AI chat	Implemented (screenshot)
FR-09	Provide sign-in, account request and administrator approval workflow	Implemented (screenshots)
FR-10	Provide a spatial map of incidents with filters	Implemented (screenshot)
FR-11	Provide a global search across crime number, accused, officer and court fields from the top bar	Implemented (search bar in screenshots); result behaviour [TO BE PROVIDED]

ID	Requirement	Status
FR-12	Provide a security-role selector offering Administrator, Police Officer, Investigation Officer, Crime Analyst and District Superintendent	Implemented (role menu in screenshot); per-role permissions [TO BE PROVIDED]
FR-13	Present an AI Risk Alerts feed and a High Risk Districts panel on the dashboard	Implemented (screenshot); alert generation method [TO BE PROVIDED]

3.6 Non-Functional Requirements

Table 4: Non-Functional Requirements

Attribute	Requirement	Status
Performance	Dashboard data returned within acceptable interactive time	Not measured [TO BE PROVIDED]
Scalability	Serverless deployment and pooled database connections	Design property; load testing [TO BE PROVIDED]
Security	Authenticated access, restricted origins, protected credentials	Partly evidenced (login and approval UI); server-side enforcement [TO BE PROVIDED]
Usability	Navigation grouped by function; filters on list and map views	Observed (screenshots)
Maintainability	Modular files (main.py, db_helper.py, groq_service.py, api.ts)	Implemented
Portability	Standard browser client; environment-based configuration	Implemented
Reliability	Error handling in API responses	Details [TO BE PROVIDED]

3.7 Hardware Requirements

The deployed system runs on Vercel's managed infrastructure, so no dedicated server hardware is required by the project. Client devices require a modern web browser. The specifications of the development machine and of the database host are [TO BE PROVIDED]; they are intentionally not estimated here.

3.8 Software Requirements

- Frontend: React [16], TypeScript and Vite [17]; Node.js toolchain (version [TO BE PROVIDED]).
- Backend: Python, Flask [18], Flask-CORS [19], Gunicorn [20] (versions [TO BE PROVIDED]).

- Database: PostgreSQL [14] with psycopg2 [21].
- AI: Groq API access key and network access to the Groq service [22].
- Deployment: Vercel account and Git/GitHub repository [23].

3.9 Feasibility Study

3.9.1 Technical feasibility

All components are established open-source or hosted technologies with public documentation [14], [16], [18], [23]. The successful production response of the KPI endpoint demonstrates that the deployed frontend-independent backend and database connection operate together.

3.9.2 Operational feasibility

The interface is grouped into Command, Users, Crime Records, Map, AI Intelligence and Operations menus, and filters are provided on list and map views, which suggests a low learning burden. No user study was conducted; usability findings are [TO BE PROVIDED].

3.9.3 Economic feasibility

The stack is built from open-source software and managed services. The Vercel dashboard screenshots show that both Vercel projects run on the Hobby plan. The PostgreSQL host and Groq plan, and any costs, are [TO BE PROVIDED]. No cost figures are asserted.

CHAPTER 4 – SYSTEM DESIGN

4.1 System Architecture

The system follows a layered client-server architecture. The user interacts with a React and TypeScript single-page frontend. The frontend calls a Flask REST API over HTTP using JSON. The Flask backend retrieves data from PostgreSQL through a database helper class and, for AI-assisted functions, calls the Groq API through a dedicated service module. Figure 1 shows the overall structure.

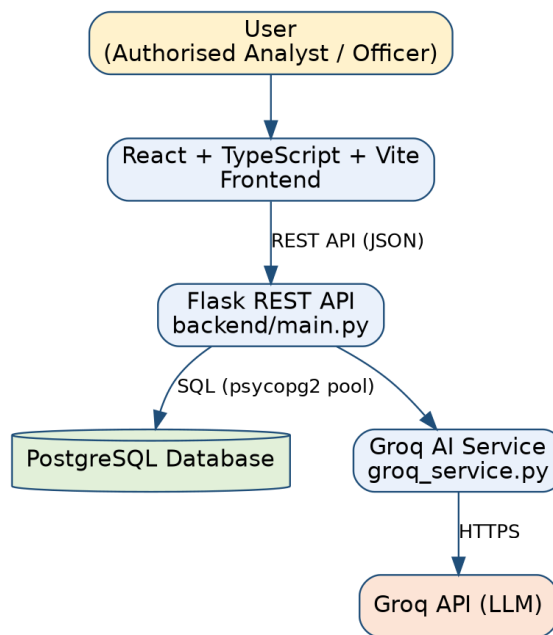


Figure 1: Overall System Architecture

4.2 High-Level Architecture

Four tiers can be distinguished: (i) the presentation tier (browser and React application); (ii) the application tier (Flask routes in backend/main.py); (iii) the data tier (PostgreSQL accessed via backend/db_helper.py); and (iv) the AI tier (backend/groq_service.py calling the external Groq API). The presentation and application tiers are deployed on Vercel; the Flask application is exposed through a Vercel Python serverless function [23]. Two Vercel projects appear in the supplied dashboard screenshots: one for the backend (ai-powered-criminal-network-analysis-system) and one for the frontend (ai-powered-criminal-network-frontend). The PostgreSQL hosting provider is [TO BE PROVIDED].

4.3 Frontend Architecture

The frontend is built with React [16] and TypeScript and bundled with Vite [17]. All HTTP communication is centralised in `frontend/src/services/api.ts`, which reads the API base address from the `VITE_API_URL` environment variable, constructs REST requests and returns parsed JSON to the page components. This design keeps endpoint addresses out of individual components and gives one place to alter base URLs between local and production environments. The navigation observed in the interface comprises Command, Users, Crime Records (FIR Management, Victim Analytics, Accused Analytics, Complainant Analytics), Map, AI Intelligence (Crime Analytics, Predictive Analytics, AI Investigation Assistant, Criminal Network) and Operations. The map and graph rendering libraries are [TO BE PROVIDED].

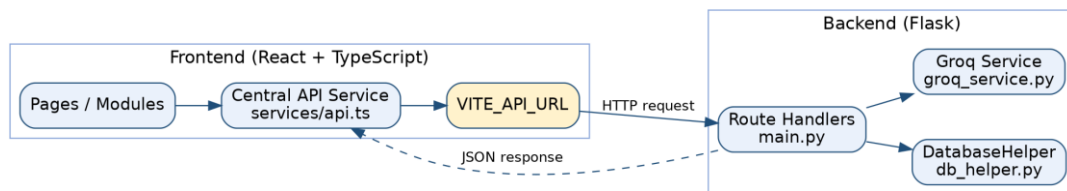


Figure 2: Frontend-Backend Architecture

4.4 Backend Architecture

The Flask application [18] is implemented in `backend/main.py`. Route handlers validate requests, delegate data access to the DatabaseHelper in `backend/db_helper.py`, delegate AI calls to `backend/groq_service.py`, and return JSON. Flask-CORS [19] adds cross-origin headers so that the browser-hosted frontend may call the API. Gunicorn [20] is the production WSGI server configured for the project; on Vercel the application is invoked as a serverless function through `api/index.py` and `backend/api/index.py` (the exact role of each entry file is [TO BE PROVIDED]).

4.5 Database Architecture

PostgreSQL [14] is the production database. The DatabaseHelper class manages connections and connection pooling, executes queries, retrieves data, inspects the schema and handles transactions. A SQLite database file is also maintained as a local backup artifact; it is not the production store. Figure 3 shows the arrangement; Chapter 5 gives the design detail.

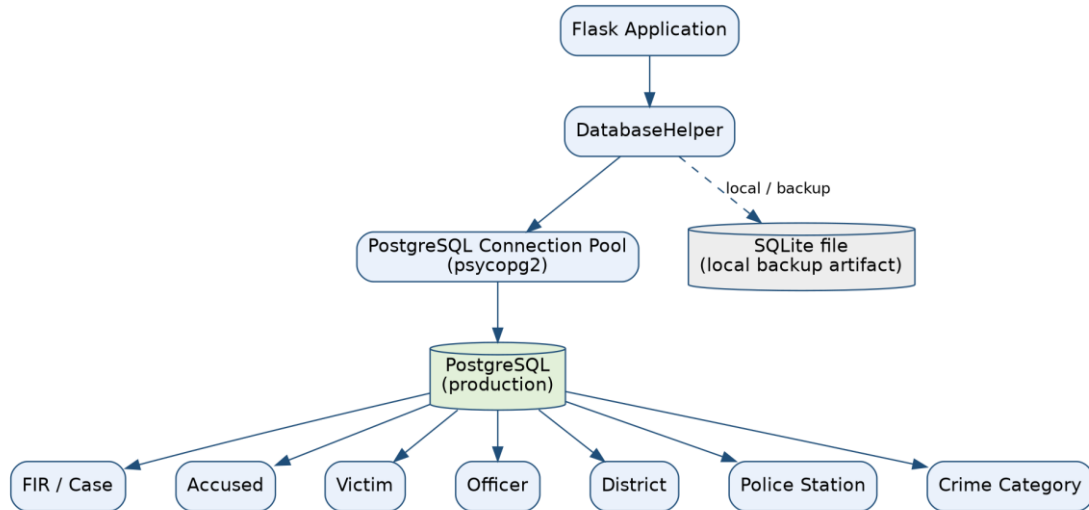


Figure 3: Database Architecture

4.6 AI Service Architecture

AI-related endpoints retrieve the relevant records, construct a context, and pass it to `groq_service.py`, which calls the Groq API [22]. The default configured model is `openai/gpt-oss-120b`, and fallback models are configured in the code (their identifiers are [TO BE PROVIDED]). The response is returned through the Flask endpoint to the frontend for analyst review. Not every endpoint uses the AI service: for example, the KPI endpoint returns database-derived counts only.

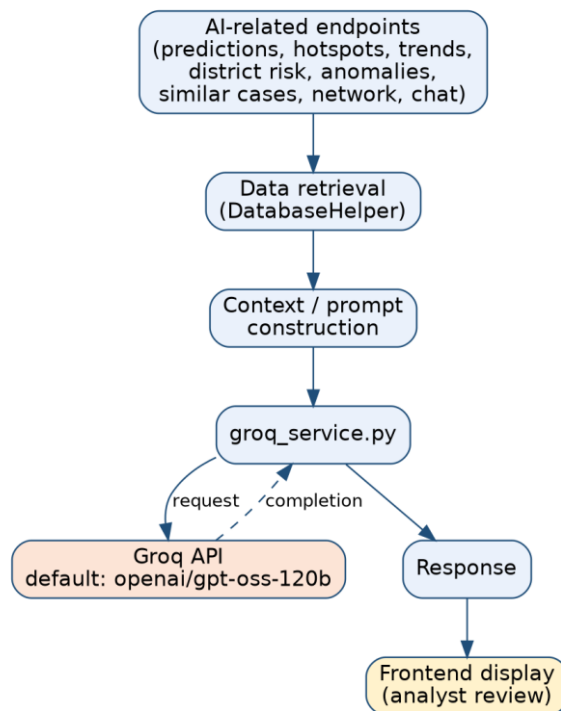


Figure 4: AI Integration Architecture

4.7 API Architecture

The API follows the REST style [15], exchanging JSON over HTTP. The verified example is GET /api/cases/kpis. The endpoint groups implemented in the backend are summarised in Table 5. Exact route paths other than the KPI route are [TO BE PROVIDED] and are listed in Appendix A.

Table 5: API Categories

Category	Purpose	Uses AI service?
Cases, case details, KPIs	Record listing, single-case view, headline counts	No (database aggregation)
Trends, districts, police stations, crime categories, demographics	Aggregated statistics by time, place, category and demographic	Not established [TO BE PROVIDED]
Officers, accused, victims	Entity-level views and summaries	Not established [TO BE PROVIDED]
AI predictions, hotspots, forecast/trends, district risk, anomalies, similar cases	Analytical and predictive views	AI-assisted (per project description); exact split [TO BE PROVIDED]
Criminal network	Entity-relationship graph data	Described as AI-derived in the interface; exact method [TO BE PROVIDED]
AI status, AI chat	Service availability and conversational queries	Yes

4.8 Data Flow

A typical request proceeds as follows: the browser component calls a function in api.ts; api.ts issues an HTTP request to the Flask route; the route calls DatabaseHelper, which obtains a pooled connection and executes SQL on PostgreSQL; rows are converted to a JSON structure and returned; the component renders the result. For AI-assisted endpoints an additional step calls groq_service.py before the response is returned. Figure 5 illustrates the KPI request.

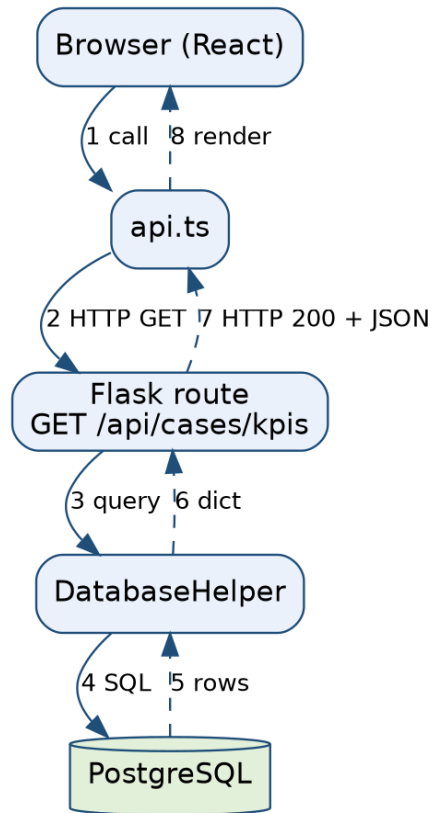


Figure 5: API Request Flow (GET /api/cases/kpis)

4.9 Module Design

The functional modules are listed in Table 6. The status column follows the labels defined in Section 1.1.

Table 6: System Modules

Module	Frontend menu / evidence	Status
Dashboard (Command Center)	Command; screenshot supplied	Implemented
Case Analytics / FIR Management	Crime Records; screenshot supplied	Implemented
Case Details	Inspect action on FIR list; endpoint listed	Implemented
KPI Analytics	KPI cards; endpoint Verified	Implemented, Verified
District and Police Station Analysis	Crime by District chart; endpoints listed	Implemented
Accused / Victim Analytics	Crime Records menu; endpoints listed	Implemented (screenshots supplied)
Officer Analysis	Officer Performance Registry; screenshot supplied	Implemented
Crime Categories and Trends	Crime Analytics; screenshot supplied	Implemented
AI Predictions, Hotspots, Anomalies	AI Intelligence Modules; screenshot supplied (hotspot tab)	Implemented

Module	Frontend menu / evidence	Status
Similar Cases, District Risk	Endpoints listed	Implemented (screens [TO BE PROVIDED])
Criminal Network	Criminal Linkage Network; screenshots supplied	Implemented
AI Chat / Investigation Assistant	AI menu; screenshot supplied	Implemented
Map	Interactive Spatial Heatmap; screenshot supplied	Implemented
User Access (login, request, approval)	Screenshots supplied	Implemented

4.10 Security Considerations

Security-relevant features evidenced in the interface are a sign-in page, an account-request form with a requested role (Police Officer, Investigation Officer, Crime Analyst, District Superintendent), and an administrator approval centre with Approve, Reject and Suspend actions and an Audit Trail link. The interface displays banner text about encryption; this report does not treat such text as evidence of a specific cryptographic implementation. The following points are considered in the design and must be confirmed against the code:

- CORS should be restricted to the deployed frontend origin rather than all origins (actual configuration [TO BE PROVIDED]).
- Server-side authentication and authorisation on every data endpoint (enforcement [TO BE PROVIDED]); front-end route hiding alone is insufficient.
- The Groq API key and database credentials must be held in server-side environment variables and never in frontend code; only non-secret values such as VITE_API_URL belong in the client bundle.
- Parameterised SQL queries should be used to prevent injection [21] (confirmation [TO BE PROVIDED]).
- Stored user passwords should be hashed (implementation [TO BE PROVIDED]).
- Crime data is personal and sensitive; access must be limited to authorised users and the data used in demonstrations should be non-identifying (Chapter 11).

4.11 System Workflows

The general workflow is: user, login, authentication, dashboard, module selection, data analysis, visualisation of results and, finally, investigator review. The workflows of the

principal modules, as observed in the interface, are as follows. Steps that were not captured are marked [TO BE PROVIDED].

- **Sign-in and role selection:** the user signs in, or submits an account request that awaits administrator approval, and then selects or switches a security role. The sign-in, request and approval screens are Implemented (Figures 6, 7 and 22); the behaviour of the interface after a role is switched was not captured [TO BE PROVIDED].
- **Dashboard:** the user opens the Command Center and views the KPI cards, the monthly trend and the AI Risk Alerts feed (Figure 10).
- **Global search:** the user enters a query in the top bar, which is labelled for crime number, accused, officer and court; the result behaviour is [TO BE PROVIDED].
- **Crime analytics:** the user opens Crime Analytics and reads the trend, district, sub-head and victim-gender charts (Figure 16).
- **FIR management:** the user opens the case list, applies filters or search, and uses the Inspect or Network action on a row (Figure 11).
- **Case network isolation:** the user selects a case in the Isolate Case Network control, the related entities are retrieved, and the connected network is displayed.
- **Suspect network isolation:** the user selects an accused person in the Isolate Suspect Network control (accused persons carry short codes such as A1 and A2), the connected entities are identified, and the suspect network is displayed; a reset control returns to the full graph (Figures 19 and 20).
- **AI assistant:** the user opens the assistant, asks a question and receives a generated response, which must be checked against the records (Figure 21).

CHAPTER 5 – DATABASE DESIGN

5.1 Database Introduction

The database holds the structured records on which every analytical view depends. Records describe cases (FIRs) and the entities associated with them. Because the same accused person, officer, station or district recurs across many cases, the data is inherently relational.

5.2 Database Selection

PostgreSQL [14] was selected as the production database for the following reasons.

- **Relational structure:** crime records are naturally expressed as related tables with keys.
- **Structured records:** fixed attributes such as registration date, status and category suit typed columns.
- **Entity relationships:** joins express accused-case-victim and case-station-district relationships directly.
- **Concurrent access:** multi-version concurrency control allows many readers alongside writers.
- **Scalability:** the server supports indexing, partitioning and replication as data grows.
- **SQL querying:** aggregate queries for KPIs, trends and distributions are concise in SQL.
- **Data integrity:** constraints and transactions protect consistency.

The project also keeps a SQLite file as a local backup artifact. PostgreSQL is the production database. The hosting provider of the production PostgreSQL instance is [TO BE PROVIDED].

5.3 Entity Relationship Concept

An entity-relationship view identifies the principal real-world objects, their attributes and the associations between them. Here the central entity is the Case (FIR); Accused, Victim, Officer, Police Station, District and Crime Category associate with it. The network screen additionally shows Court nodes attached to cases; whether courts are stored as a separate entity is [TO BE PROVIDED]. The full set of entities documented for the network view is Case, FIR, Accused, Victim, Complainant, Officer, Police Station, District, Court, Crime

Head and Legal Section. Complainant records are named in the module list (Complainant Analytics) but had no distinct node colour in the captured network screens, so their relationship to Case is [TO BE PROVIDED].

5.4 Entity Relationships

Figure 23 (Chapter 8) depicts the principal relationship types displayed by the network module: ACCUSED_IN, VICTIM_IN, CLASSIFIED_AS, REGISTERED_AT and TRIED_IN. The interface also labels INVESTIGATED (officer and case), IN_DISTRICT (station and district), CONTAINS_SECTION (crime head and legal section) and CHARGED_UNDER (case and legal section); the last links a case directly to a legal section in addition to the section reached through its crime head, and should be confirmed against the schema. Conceptually, one accused may be linked to many cases; a case may have several accused and several victims; a case is registered at one police station, which belongs to one district; and a case is classified under a crime category. A case is also tried in a court and investigated by an officer, and falls under one or more legal sections; the exact cardinalities of these relationships are [TO BE PROVIDED].

5.5 Database Schema

[EXACT DATABASE SCHEMA TO BE INSERTED FROM DATABASE IMPLEMENTATION]

Table and column names are not reproduced here because they were not supplied and must not be invented. The DatabaseHelper provides schema inspection, so the authoritative schema can be exported from PostgreSQL (for example with the `psql \d` command or `pg_dump --schema-only`) and inserted in Appendix C.

5.6 Important Entities

Table 7: Database Entities (Conceptual)

Entity	Conceptual description	Related to
FIR / Case	A registered case with a case number, registration date, status, crime category and station	Accused, Victim, Officer, Police Station, Crime Category
Accused	A person associated with one or more cases	Case
Victim	A person recorded as a victim in a case	Case
Officer	An investigating officer identified in the case list	Case, Police Station
District	An administrative region containing stations	Police Station

Entity	Conceptual description	Related to
Police Station	A station where cases are registered	Case, District, Officer
Crime Category	Major head and sub-head classification of a case	Case
Court	A court to which a case is attached in the network view (a JMFC court and a District & Sessions court appear in the captured graph)	Case
Legal Section	A legal section (shown, for example, as Sec. 379) reached through a crime head and, by the CHARGED_UNDER label, directly from a case	Crime Category, Case
Complainant	A person who reports a case; named in the Complainant Analytics module, but representation in the graph is [TO BE PROVIDED]	Case (to be confirmed)
User account	Portal user with requested role and approval status	(access control)

Exact attributes are [EXACT DATABASE SCHEMA TO BE INSERTED FROM DATABASE IMPLEMENTATION].

5.7 Relationships

The cardinalities stated in Section 5.4 are conceptual and are inferred from the relationships displayed in the interface; they should be checked against the foreign keys in the implemented schema [TO BE PROVIDED].

5.8 Data Integrity

PostgreSQL enforces integrity through primary keys, foreign keys, NOT NULL and CHECK constraints, and ACID transactions [14]. The DatabaseHelper handles transactions. Which constraints are declared in the deployed schema is [TO BE PROVIDED]. The consistency of the verified statistics is an example of integrity at the data level: the solved-case count (110) equals the sum of charge-sheeted (51) and closed (59) cases, and the interface reports a resolution rate of 92 percent, which corresponds to 110 of 120 cases. These observations describe the data and are not a substitute for constraint verification.

5.9 Connection Pooling

Opening a database connection is costly. A connection pool keeps a set of open connections that request handlers borrow and return, reducing per-request latency and bounding the number of simultaneous connections to the server. The DatabaseHelper uses PostgreSQL connection pooling with psycopg2 [21]. In a serverless deployment each function instance

maintains its own pool, so pool sizes must be chosen with the database connection limit in mind. The configured pool size is [TO BE PROVIDED].

5.10 Database Security

- Credentials must be supplied through environment variables, not source code.
- Connections to a hosted database should use TLS (configuration [TO BE PROVIDED]).
- The application database role should have least privilege.
- Backups and the SQLite artifact contain the same sensitive records and must be protected accordingly.

CHAPTER 6 – IMPLEMENTATION

6.1 Implementation Overview

The repository is organised into a frontend application and a backend application. The files central to this chapter are `frontend/src/services/api.ts`, `backend/main.py`, `backend/db_helper.py`, `backend/groq_service.py`, `backend/api/index.py` and `api/index.py`. Only relevant fragments are described; source listings should be taken from the repository (Appendix B).

6.2 Frontend Implementation

The frontend is a Vite-built React application written in TypeScript [16], [17]. Pages render KPI cards, an area chart of monthly crime volume, bar charts of crime by district, a horizontal bar chart of crime sub-heads, a pie chart of victim gender, a map, a relationship graph and tables. The chart library is [TO BE PROVIDED]. Sign-in and account-request pages (Figure 6 and Figure 7) precede the portal.

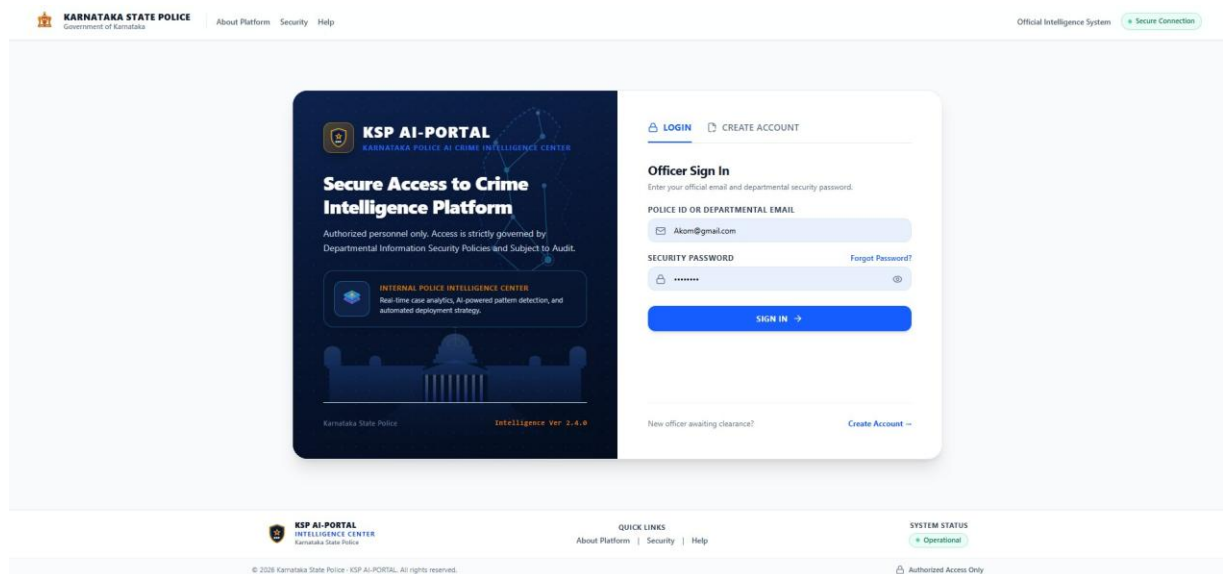


Figure 6: Sign-in Page

The screenshot shows a web form for requesting access to the KSP AI-PORTAL. The form is titled 'KSP AI-PORTAL' and 'INTELLIGENCE CENTER CREDENTIAL PROVISIONING'. It includes the following fields and options:

- FULL NAME ***: A text input field with a placeholder 'Enter full name' and a red error message 'Full Name is required.' below it.
- POLICE / EMPLOYEE ID ***: A text input field with a placeholder 'e.g. KSP-821-4567' and a red error message 'Police / Employee ID is required.' below it.
- SECURITY PASSWORD ***: A password input field with a red error message 'For better password' below it.
- REQUESTED ROLE ***: A dropdown menu with the following options: 'Police Officer', 'Investigation Officer', 'Crime Analyst', and 'District Superintendent'. The 'Police Officer' option is currently selected.
- CONFIRM PASSWORD ***: A password input field with a red error message 'For better password' below it.
- Submit Access Request**: A blue button at the bottom right.

At the bottom of the form, there is a link 'Already have an authorized account? Sign In'.

Figure 7: Account Request Form with Requested Role Selection

6.3 Backend Implementation

backend/main.py creates the Flask application, enables CORS and registers route handlers grouped by the categories in Table 5. Each handler obtains data through DatabaseHelper or through the AI service and returns a JSON response. Gunicorn serves the application in production [20].

6.4 REST API Implementation

Endpoints are exposed under the /api prefix (for example /api/cases/kpis). Requests are HTTP calls with JSON responses. The complete list of routes is [TO BE PROVIDED] (Appendix A).

6.5 Database Integration

backend/db_helper.py defines the DatabaseHelper class, which establishes the PostgreSQL connection pool, executes queries, retrieves data, inspects the schema and manages transactions [21]. Handlers depend on this class rather than on driver calls, which centralises connection handling.

6.6 AI Integration

backend/groq_service.py wraps calls to the Groq API [22]. The default model is openai/gpt-oss-120b and fallback models are configured, so that a failure or unavailability of the primary model can be handled by another configured model (the fallback policy detail is [TO BE PROVIDED]). An AI status endpoint reports the availability of the AI service.

6.7 CORS Configuration

Browsers block cross-origin requests unless the server sends appropriate headers. Flask-CORS [19] supplies these headers so that the frontend origin may call the backend origin. The allowed-origins setting used in production is [TO BE PROVIDED].

6.8 Error Handling

The backend is expected to return an HTTP error status and a JSON error body when a database or AI call fails, and the frontend API service handles non-successful responses. The concrete implementation of error handling is [TO BE PROVIDED].

6.9 API Request/Response Flow

The verified production exchange is a GET request to `/api/cases/kpis`, which returned HTTP 200 and the following JSON body:

```
{
  "charge_sheeted": 51,
  "closed_cases": 59,
  "pending_cases": 10,
  "solved_cases": 110,
  "today_firs": 1,
  "total_accused": 79,
  "total_districts": 7,
  "total_firs": 120,
  "total_stations": 29,
  "total_victims": 120
}
```

The frontend receives this JSON through `api.ts` and displays it in the dashboard cards (Figure 10). The same values appear on the dashboard screenshot: 120 total FIRs, 110 solved cases (51 charge-sheeted and 59 closed), 79 accused, and 29 stations across 7 districts.

6.10 Deployment Implementation

The application is deployed on Vercel [23]. The Flask application is exposed as a Python serverless function through `api/index.py` (with `backend/api/index.py` as a corresponding entry). The frontend obtains the backend address from `VITE_API_URL`. The Vercel dashboard for the backend project shows a production deployment with status Ready, built from the main branch at commit 95975cf ("Fix Vite base path for Vercel deployment"), on the Hobby plan; over the observed 6-hour window it recorded 162 CDN requests, 162 function invocations and a 0 percent error rate (Figure 8). The frontend project shows a

Ready production deployment of the same commit whose preview is the sign-in page, with 20 CDN requests and 0 function invocations in the same window (Figure 9). The frontend serves static assets and issues its API calls to the backend project. Environment-variable values are not shown and must not be printed in the report; the production domain names visible in the screenshots may be quoted if the department permits.

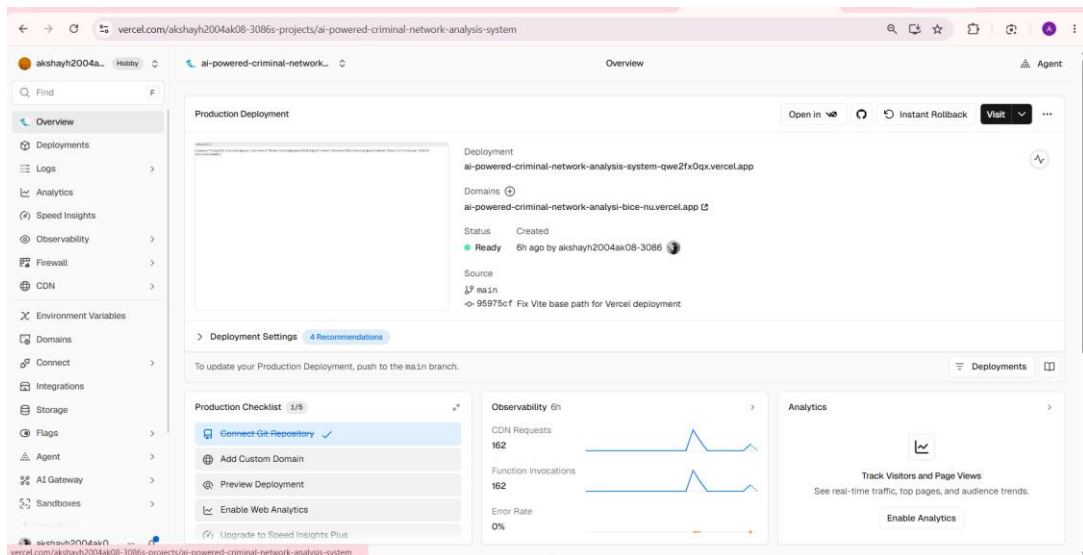


Figure 8: Production Deployment: Backend Project on Vercel

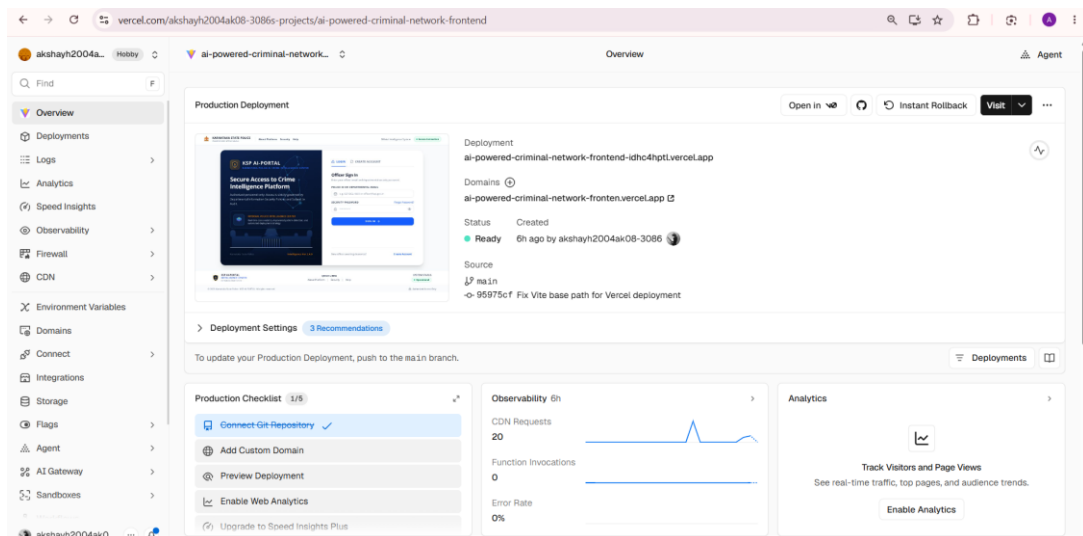


Figure 9: Production Deployment: Frontend Project on Vercel

CHAPTER 7 – MODULES

Each module is described by purpose, input, processing and output, and user benefit. Where the exact backend computation was not supplied, this is stated rather than assumed.

7.1 Dashboard (Command Center)

Status: Implemented (screenshot supplied).

Purpose. Give an at-a-glance overview of recorded crime.

Input. KPI data from the KPI endpoint, monthly counts, and AI-related alerts and district information shown on the page.

Processing. The frontend requests the data through `api.ts` and renders KPI cards, an area chart of monthly crime volume, an "AI Risk Alerts" panel and a "High Risk Districts" section. Whether the alert texts are generated live or are sample content is [TO BE PROVIDED]. The alerts visible in the supplied capture report a high-probability spike in cyber fraud in the Bengaluru district, a repeat offender (A2) detected in the Devaraja PS area, and an active burglary gang pattern across Mysuru stations. The panel carries a LIVE label and severity-styled icons. Each alert is an analytical lead for human review and not a finding. The top bar also holds the global search box, a system-status badge, a notification bell and a user and role indicator, and the sidebar shows a clearance-level indicator; none of these is evidence of server-side access control (Section 4.10).

Output. Total FIRs 120, solved cases 110, active suspects 79 and 29 stations in 7 districts (Figure 10).

User benefit. A single entry view of case load and resolution status.

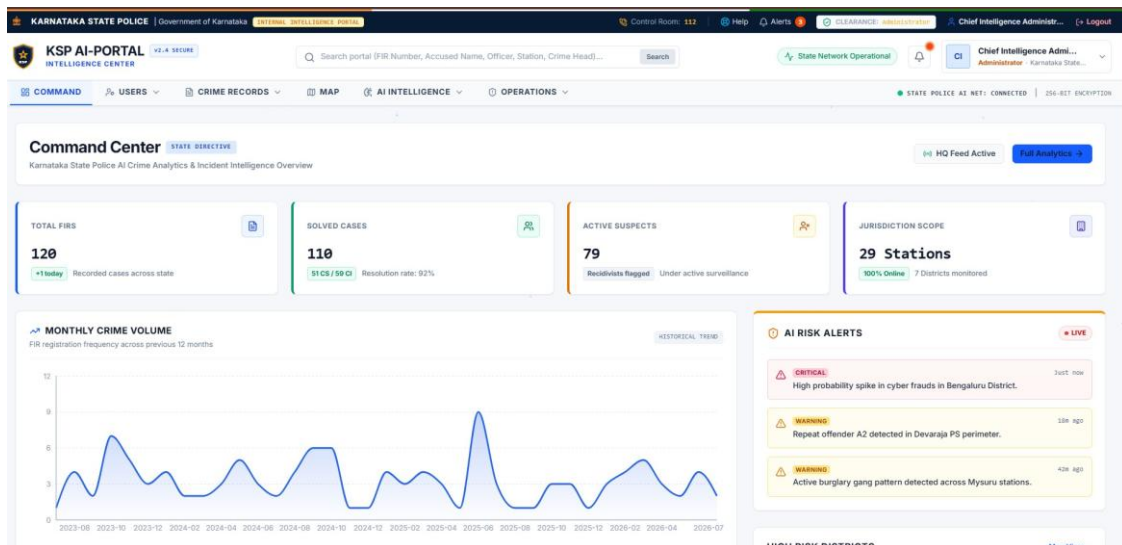


Figure 10: Dashboard (Command Center)

7.2 Case Analytics (FIR Management)

Status: Implemented (screenshot supplied).

Purpose. List and search registered cases.

Input. Search text and filters for district, crime category and case status.

Processing. The frontend requests case records from the cases endpoint; the table shows crime/case number, registration date, major head and sub-head, police station, investigating officer and status, with Inspect and Network actions per row.

Output. A filtered case table (Figure 11).

User benefit. Quick location of a case and direct navigation to its details or its network.

CRIME / CASE NO	REG DATE	MAJOR HEAD	POLICE STATION	INVESTIGATING OFFICER	STATUS	ACTIONS
100030018202600046 Case: 202600046	2026-07-22	Crimes Against Public Order Rioting	Vidyanagar PS	PC Raghu Gowda	Under Investigation	Inspect Network
300010007202600037 Case: 202600037	2026-07-10	Property Crimes Burglary	Majestic PS	Insp. Anil Pujar	Under Investigation	Inspect Network
200040020202600069 Case: 202600069	2026-06-26	Cyber Crimes Online Fraud	Pandeshwar PS	PC Kiran Patil	Charge Sheeted	Inspect Network
100040023202600025 Case: 202600025	2026-06-12	Property Crimes Robbery	Bunder PS	SI Kiran Patil	Under Investigation	Inspect Network
100050027202600034 Case: 202600034	2026-06-02	Crimes Against Body Murder	APMC PS	SI Raghu Mathad	Charge Sheeted	Inspect Network
100040021202600035 Case: 202600035	2026-05-21	Cyber Crimes	Wardhi DC	SI Sandhu Dhanraj	Charge Sheeted	Inspect Network

Figure 11: FIR Records Management with Crime Records Menu

7.3 Case Details

Status: Implemented (endpoint listed; Inspect action shown on the case list).

Purpose. Show the complete record of one case.

Input. A case identifier chosen from the case list.

Processing. A case-details endpoint retrieves the record and associated entities. Displayed fields are [TO BE PROVIDED].

Output. A detailed case view.

User benefit. Examination of one case without leaving the portal.

7.4 KPI Analytics

Status: Implemented; the endpoint is Verified.

Purpose. Provide headline counts.

Input. A GET request to /api/cases/kpis with no parameters.

Processing. The backend computes counts from the database and returns a JSON object.

Output. Ten values: total FIRs 120, solved 110, closed 59, pending 10, charge-sheeted 51, accused 79, victims 120, districts 7, stations 29, today's FIRs 1. The dashboard cards present them as Total FIRs, Solved Cases (51 CS / 59 CI, resolution rate 92%), Active Suspects (79) and Jurisdiction Scope (29 stations, 7 districts) (Figure 12). The card labelled Active Suspects displays the total_accused value; the phrase "under active surveillance" is interface text and is not a verified statement.

User benefit. Reliable summary numbers for management reporting. These are database statistics and not model-performance measures.

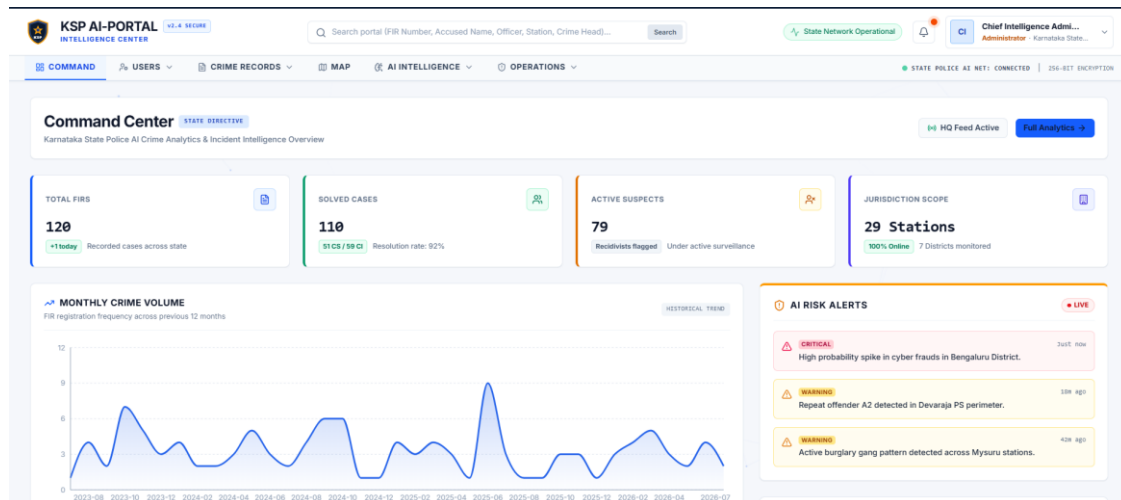


Figure 12: KPI Analytics: Dashboard KPI Cards and Monthly Crime Volume

7.5 District Analysis

Status: Implemented (chart in screenshot; endpoint listed).

Purpose. Compare crime volume by district.

Input. District data derived from case records.

Processing. Cases are aggregated by district. The seven districts shown are Kalaburagi, Belagavi, Hubballi-Dharwad, Mangaluru, Shimoga, Bengaluru City and Mysuru.

Output. A bar chart in which Bengaluru City has the largest bar (Figure 16).

User benefit. Identification of districts with higher recorded volume.

7.6 Police Station Analysis

Status: Implemented (endpoint listed; station appears in case, hotspot and officer views).

Purpose. Analyse cases and officers by station.

Input. Station identifiers from case records.

Processing. Aggregation of cases per station; 29 stations are recorded. Detailed station screen is [TO BE PROVIDED].

Output. Station-level counts and rankings.

User benefit. Support for station-level resource questions.

7.7 Accused Analysis

Status: Implemented (screenshot supplied).

Purpose. Profile accused persons and their case involvement.

Input. Accused records, case links.

Processing. Retrieval of accused records and their associated cases. The page "Accused & Repeat Offender Analytics" shows 79 suspect profiles and 12 high-risk repeat offenders flagged, and a registry listing accused ID, name, number of cases, number of arrests and a repeat score with a View Details action (Figure 13). Many entries display the maximum score of 100, which suggests a capped or saturating score; the scoring formula and the criterion for the 12 flagged suspects are [TO BE PROVIDED]. The score is an internal indicator and not evidence of guilt.

Output. An accused-level registry.

User benefit. Ability to see repeated involvement across cases.

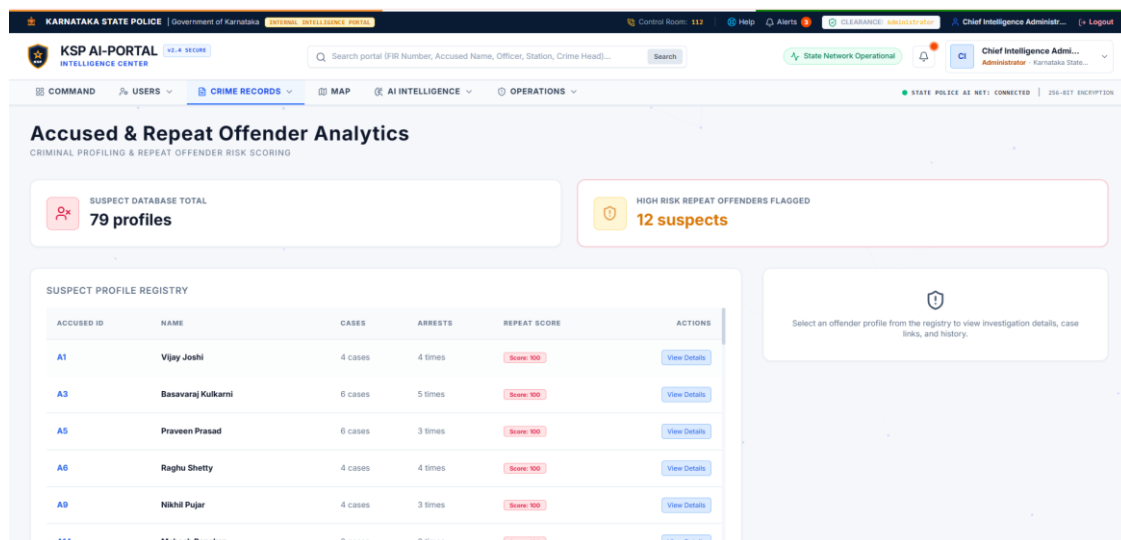


Figure 13: Accused and Repeat Offender Analytics

7.8 Victim Analysis

Status: Implemented (screenshots supplied).

Purpose. Summarise victim demographics.

Input. Victim records linked to cases.

Processing. Aggregation of victims by demographic attributes. The Victim Analytics page shows 120 total victims registered, 23 police personnel injured or affected, and a registry with name, age, gender, a police-personnel flag and case involvement (Figure 14). The gender chart in Figure 16 shows 77 percent male and 23 percent female among recorded victims. Two observations need checking against the data: the card "Female victims assisted" shows 0, and the case-involvement column shows only the word "cases" without a count. Both are [TO BE PROVIDED / VERIFIED] before final submission.

Output. Victim registry and distribution charts.

User benefit. Understanding of who is affected by recorded crime.

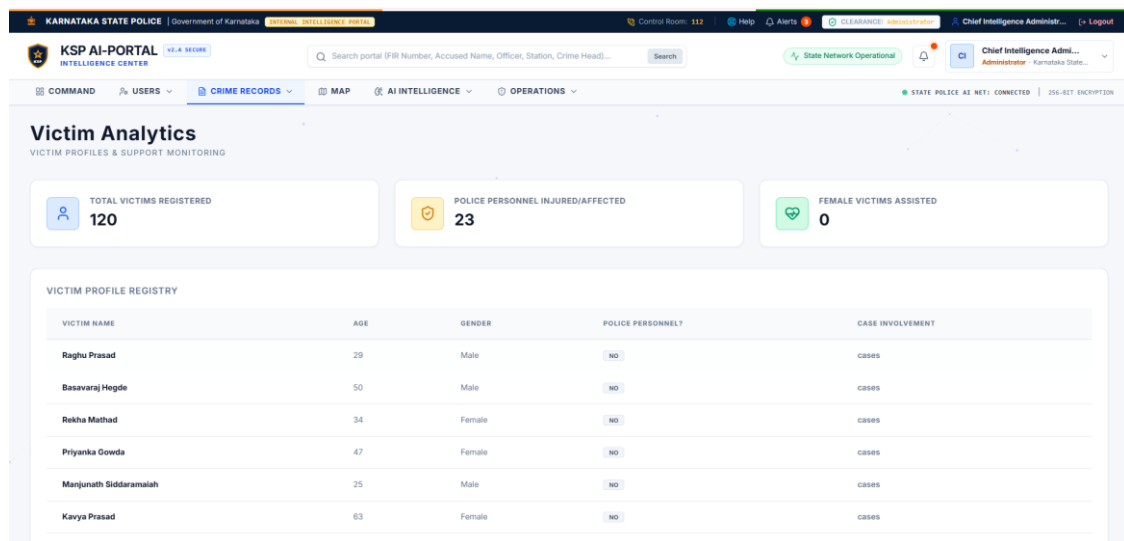


Figure 14: Victim Analytics

7.9 Officer Analysis

Status: Implemented (screenshot supplied).

Purpose. Show investigating-officer workload and resolution.

Input. Officer, station and case-status data.

Processing. For each officer the table lists registered, solved and pending cases and a resolution rate. The page reports 91 active investigating officers and an average resolution rate of 91 percent (Figure 15). These are descriptive workload figures and should not be used as a measure of individual competence without context.

Output. An officer resolution roster.

User benefit. Visibility of workload distribution.

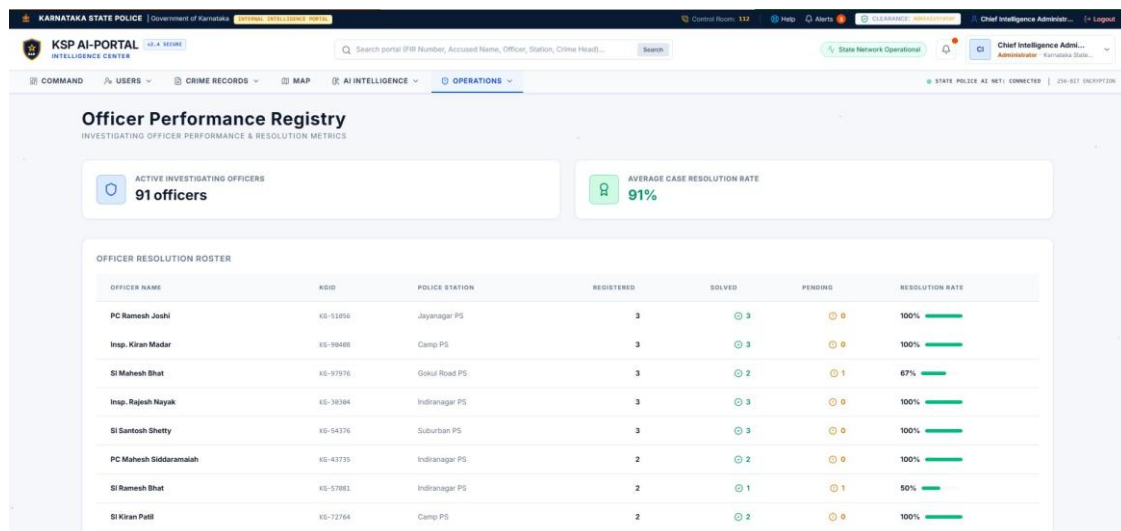


Figure 15: Officer Performance Registry

7.10 Crime Categories

Status: Implemented (screenshot supplied).

Purpose. Show the distribution of crime types.

Input. Crime major head and sub-head fields.

Processing. Aggregation by sub-head; the chart in Figure 16 lists categories such as Murder, Online Fraud, Drug Trafficking, Rioting, Hacking, Theft, Assault, Kidnapping, Burglary and Robbery.

Output. A horizontal bar chart.

User benefit. Insight into the mix of recorded offences.

7.11 Crime Trends

Status: Implemented (screenshot supplied).

Purpose. Show change over time.

Input. Case registration dates.

Processing. Cases are counted per month over the recorded period (the axes span 2023-08 to 2026-07) and drawn as a line/area chart.

Output. A monthly trend chart.

User benefit. Detection of seasonal patterns and spikes in recorded volume.

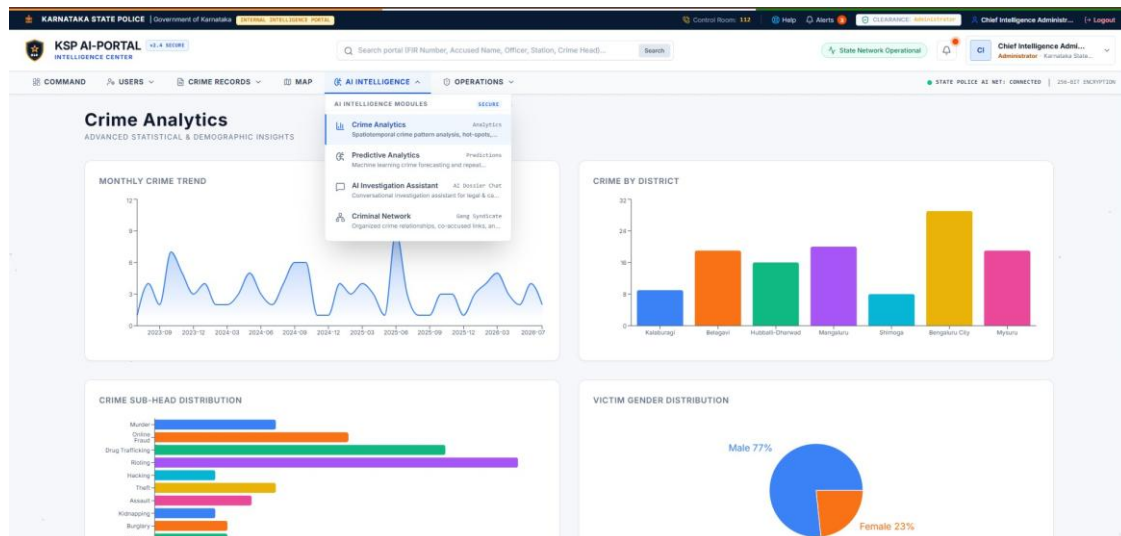


Figure 16: Crime Analytics: Monthly Trend, District, Sub-head and Victim Gender Charts

7.12 AI Predictions

Status: Implemented (AI Intelligence Modules page supplied).

Purpose. Provide predictive and pattern-discovery views.

Input. Case data and the AI service.

Processing. The page contains three tabs: Crime Hotspot Prediction (28 items), Crime Trend Forecast (labelled "increasing") and Anomaly Detection (7 items), with a Refresh Predictions control. The prediction method behind each tab is [TO BE PROVIDED].

Output. Ranked items with risk scores and suggested actions (Figure 17).

User benefit. A prioritised set of areas for analyst attention.

7.13 Hotspot Analysis

Status: Implemented (screenshots supplied).

Purpose. Identify areas with concentrated recorded incidents.

Input. Incident counts, heinous-crime ratio, recency and coordinates.

Processing. The interface states that items are ranked by volume, heinous ratio and recency. Each item shows incidents, recent incidents, trend, top crime type, coordinates, a risk score out of 100 and an AI analysis note with a stated confidence. The exact scoring formula and the meaning of "confidence" are [TO BE PROVIDED]; these values are not validated accuracy figures. A separate map plots incidents with district, crime head and status filters (Figure 18).

Output. Ranked hotspots and a map.

User benefit. Focus on places with concentrated activity, consistent with hot-spot criminology [11].

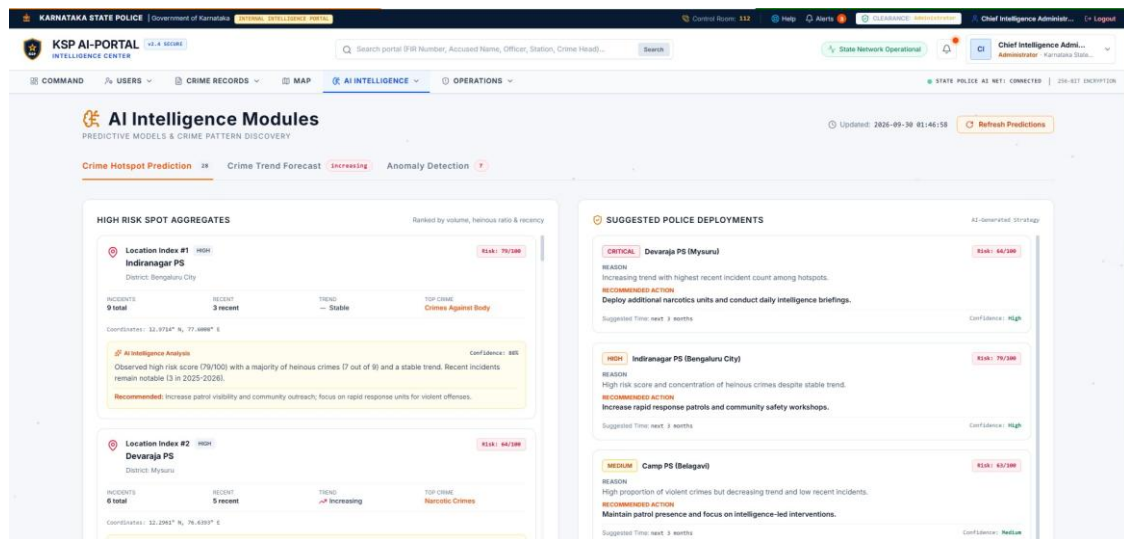


Figure 17: AI Intelligence Modules: Crime Hotspot Prediction

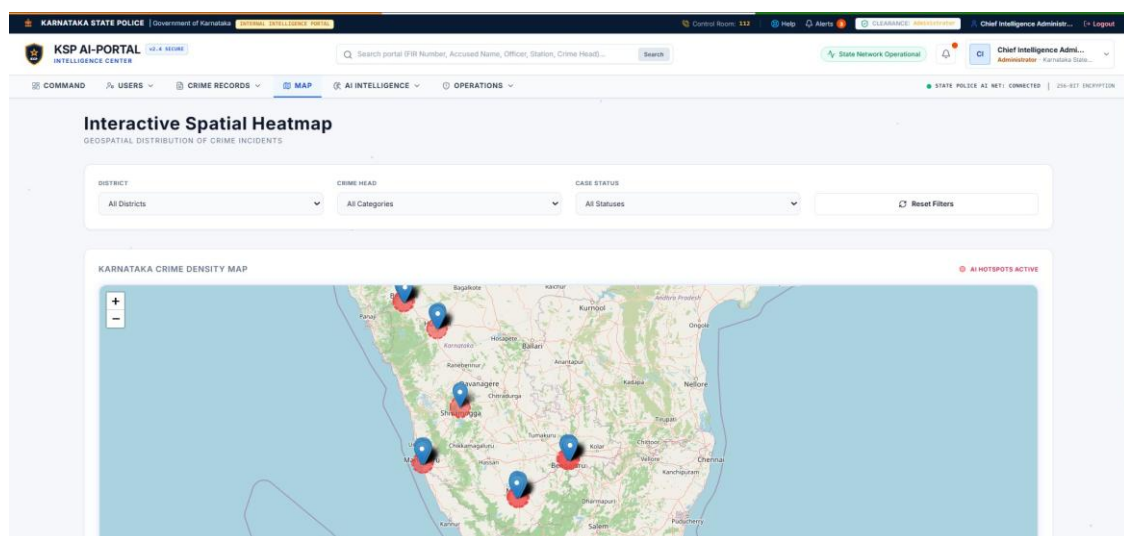


Figure 18: Interactive Spatial Heatmap of Crime Incidents

7.14 Anomaly Analysis

Status: Implemented (tab and endpoint listed; tab content screenshot [TO BE PROVIDED]).

Purpose. Flag unusual patterns.

Input. Case data.

Processing. The tab reports 7 anomalies. The detection technique is [TO BE PROVIDED]; anomaly types are discussed conceptually in [13].

Output. A list of flagged items.

User benefit. Prompts analysts to examine departures from expected patterns.

7.15 Similar Case Analysis

Status: Implemented (endpoint listed; screen [TO BE PROVIDED]).

Purpose. Find cases resembling a given case.

Input. A case identifier.

Processing. The similarity method (attribute matching, embedding or AI-based comparison) is [TO BE PROVIDED].

Output. A list of related cases.

User benefit. Helps link possibly related incidents for human review.

7.16 Criminal Network Analysis

Status: Implemented (screenshots supplied).

Purpose. Visualise relationships among accused, victims, cases, crime heads, stations and courts.

Input. Optional selection of a case or of an accused person; otherwise the global network.

Processing. The network endpoint returns nodes and typed edges. Node types are accused, victim, officer, case, crime head, court and station; edge labels include ACCUSED_IN, VICTIM_IN, CLASSIFIED_AS, REGISTERED_AT and TRIED_IN. Selecting an accused person (for example the entries labelled A42 and A1 in the selector) isolates that person's neighbourhood (Figure 19 and Figure 20). An inspector panel shows details of a clicked node. The page is captioned "AI-derived relationship and modus operandi graph"; the extent to which the AI service contributes to edge construction is [TO BE PROVIDED]. Centrality, community detection and graph neural networks are not implemented.

The page provides an Isolate Case Network dropdown, an Isolate Suspect Network dropdown and a reset control, and accused persons are identified by short codes such as A1 and A2. Besides the node types listed above, district and legal-section nodes appear, and further edge labels are INVESTIGATED, IN_DISTRICT, CONTAINS_SECTION and CHARGED_UNDER (Section 5.4). In its initial state the Inspector Panel instructs the user to click any node to view relational profile facts, details and linkages; the fields shown for a selected node are [TO BE PROVIDED].

The on-screen legend colour-codes four node types explicitly: accused (red or pink), victim (green), officer (blue) and case (cyan). The colours of the remaining types were read from the graph itself: court (orange or amber), district and station (purple), legal section (grey) and crime head (teal or green).

Output. An interactive node-link graph.

User benefit. Reveals repeated involvement and shared context without proving guilt.

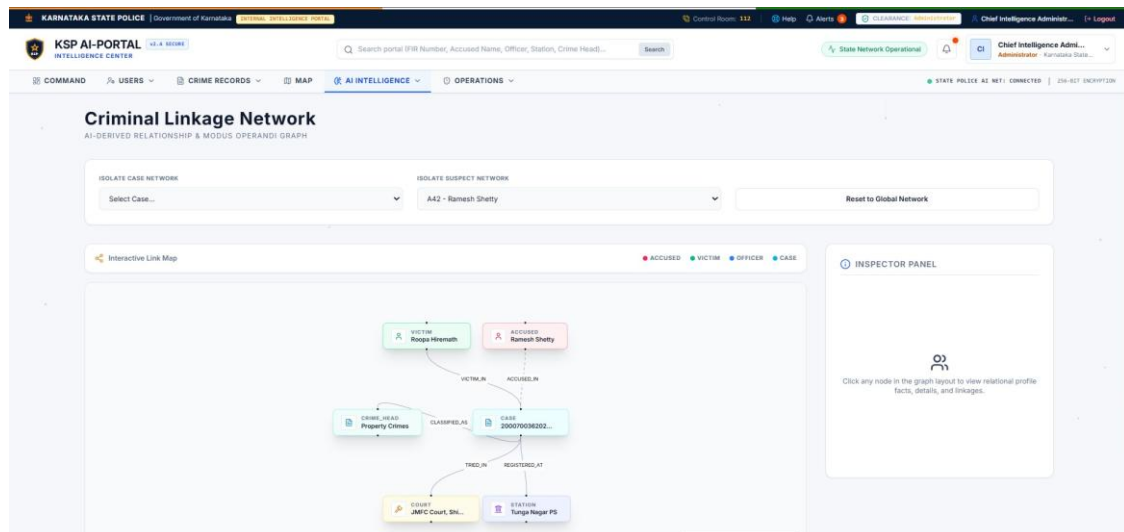


Figure 19: Criminal Linkage Network: Isolated Suspect Network

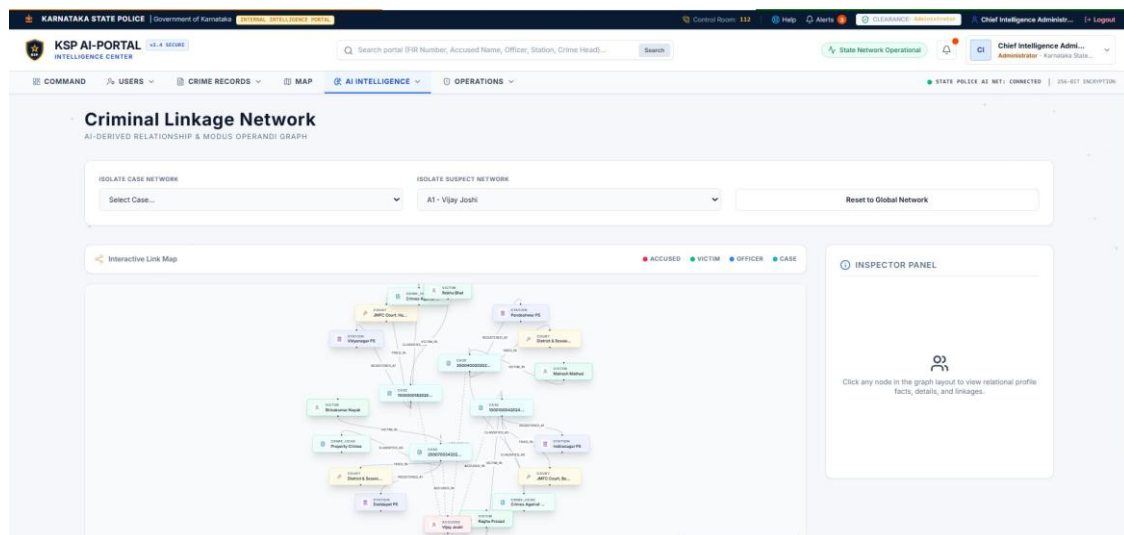


Figure 20: Criminal Linkage Network: Suspect with Multiple Linked Cases

7.17 AI Chat

Status: Implemented (screenshot supplied).

Purpose. Answer natural-language questions about the data.

Input. A user question.

Processing. The backend forwards the question, with relevant context, to the Groq model through `groq_service.py`. The page shows an "AI Online – Groq" status indicator, a greeting reporting "120 records verified", quick-question shortcuts (Crime Trends, Most Common Crimes, Location Analysis, Recent Crime Data, Crime Statistics, Explain Dashboard) and sample questions. For the question about the most common crime categories, the reply is headed "Database Fact", cites 120 case files dated 2023-07-28 to 2026-07-22, and lists Crimes Against Public Order (30 cases, 25.0%), Narcotic Crimes (24, 20.0%) and Crimes Against Body (23, 19.2%) (Figure 21). This indicates that database-derived data is supplied to the model; the retrieval mechanism is [TO BE PROVIDED]; retrieval grounding is discussed in [9]. Chat output remains generated text and must be checked against the records.

Output. A natural-language response.

User benefit. Conversational access to insights, subject to verification.

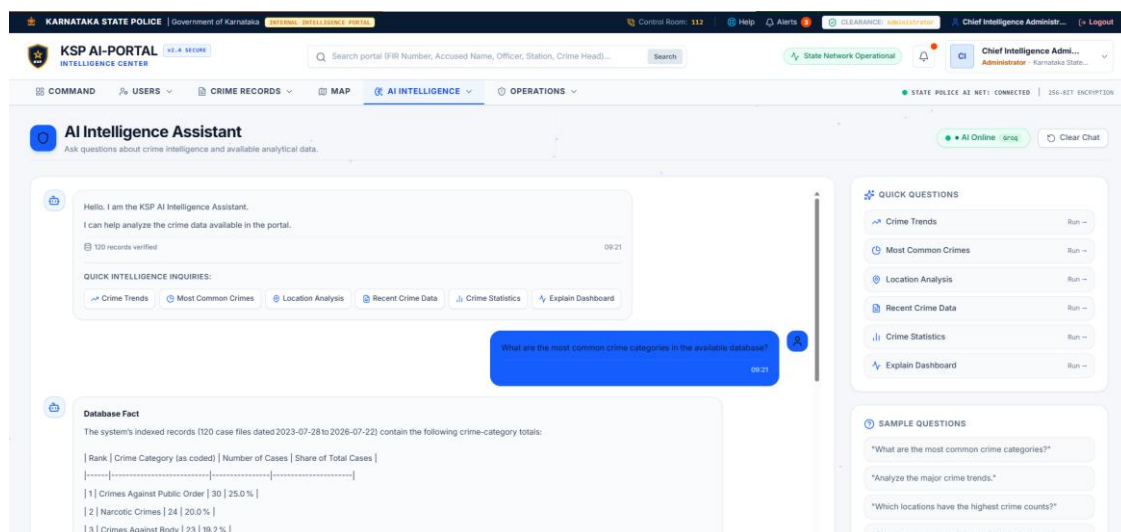


Figure 21: AI Intelligence Assistant (Chat)

7.18 Supporting Module: User Access and Approval

Although not in the requested module list, the screenshots show a User Approval Center (Figure 22) with counts of pending (1), approved (4), suspended (0) and rejected (0) users, a filterable user table with role and district and department identifiers, and Approve, Reject and Suspend actions. This indicates an implemented registration-and-approval workflow. Server-side permission enforcement by role is [TO BE PROVIDED]. Fine-grained role-based access control and full audit logging are listed as future work in Chapter 12 because their completeness could not be established.

KARNATAKA STATE POLICE

Government of Karnataka

INTERNAL INTELLIGENCE PORTAL

Control Room: 112

Help

Alerts

Chief Intelligence Administrator

Chief Intelligence Administrator

Logout

KSP AI-PORTAL

Version: 4.0

Secure

Search portal (SIR Number, Accused Name, Officer, Station, Crime Head)...

Search

State Network Operational

Chief Intelligence Administrator

Karnataka State...

COMMAND

USERS

CRIME RECORDS

MAP

AI INTELLIGENCE

OPERATIONS

STATE POLICE AT NET: CONNECTED

256-BIT ENCRYPTION

User Approval Center

Official Credential Verification & Law Enforcement Role Access Control

Users Directory

Audit Trail

Pending Approval

1

Requires administrator review

Active & Approved

4

Full authorized portal access

Suspended

0

Temporary clearance revocation

Rejected

0

Denied registration requests

Search by name, email, badge ID, district...

All Records

Pending

Approved

Suspended

Rejected

USER	NAME / OFFICIAL EMAIL	ROLE / CLEARANCE	DISTRICT	DEPARTMENT ID	STATUS	REGISTERED	ACTIONS
A	akshay ca18@ksp.gov.in	Police Officer	Karnataka State Police HQ	ca01	Approved	9/29/2026	Suspend
O	omkar ksp02@ksp.gov.in	Police Officer	Karnataka State Police HQ	ksp02	Approved	9/29/2026	Suspend
O	omkar ksp03@ksp.gov.in	Police Officer	Karnataka State Police HQ	ksp03	Pending	9/29/2026	Approve Reject
P	POLICE1 1234@ksp.gov.in	Police Officer	Karnataka State Police HQ	1234	Approved	9/29/2026	Suspend
CI	Chief Intelligence Administrator ciadmin@ksp.gov.in	Police Officer	Karnataka State Police HQ	KSP-IHQ-001	Approved	9/29/2026	

Figure 22: User Approval Center

CHAPTER 8 – AI AND ANALYTICS

8.1 AI Architecture

The AI layer is an intelligence-assistance layer between the data and the analyst. It is not a trained model developed in this project. Data are retrieved by Flask handlers, formulated into requests, sent to a hosted language model via the Groq API, and returned for display. No model training, fine-tuning or benchmark evaluation was performed in this project. Not all endpoints use the AI layer; descriptive analytics such as KPIs are computed directly from the database.

8.2 Groq API Integration

`groq_service.py` encapsulates access to the Groq API [22]. The default model is `openai/gpt-oss-120b`, and fallback models are configured in the code. An AI status endpoint allows the frontend to check whether the AI service is available. API keys are provided through server-side configuration (to be confirmed, Section 4.10).

8.3 AI Chat

The chat function accepts a natural-language query and returns a generated response. In the supplied screenshot the reply is labelled "Database Fact" and its category totals (30, 24 and 23 cases) are consistent with the 120 recorded cases. Because language models can produce unsupported statements [7], chat responses should be presented as suggestions, cross-checked against the underlying records. Grounding by retrieval [9] is recommended and is discussed as future work.

8.4 Crime Trend Analysis

Trend views aggregate case counts by month. The forecast tab is labelled "increasing" in the interface. The forecasting technique and any validation are [TO BE PROVIDED]. Statistical crime-forecasting literature [8], [12] illustrates what rigorous evaluation would involve, but this project reports no forecast error metrics.

8.5 Hotspot Analysis

Hotspots are ranked using volume, the proportion of heinous crimes and recency, as stated on the interface. Each hotspot receives a risk score out of 100 and a textual analysis. With

only 120 records in seven districts, per-station counts are small (for example nine incidents for the top-ranked station in Figure 17), so scores are sensitive to single cases and must be read as indicative.

8.6 Anomaly Analysis

Anomaly analysis flags items that depart from expected patterns [13]. Seven items were flagged in the supplied screenshot. Whether an item reflects a real irregularity, a data-entry artefact or a small-sample effect requires human review.

8.7 Similar Case Analysis

Similar-case analysis retrieves cases that share attributes or narrative features with a selected case. Its purpose is to prompt investigators to consider linkages; similarity does not imply common authorship of offences.

8.8 District Risk Analysis

District risk analysis summarises risk at district level. The dashboard shows a High Risk Districts panel. The risk definition is [TO BE PROVIDED]. Risk labels for districts can stigmatise areas if misread, so they should be accompanied by the underlying counts.

8.9 Criminal Network Analysis

A criminal network is a graph in which nodes are entities and edges are relationships [4], [5]. In this system the nodes are accused persons, victims, officers, cases, crime heads, courts and stations, and edges are typed relations. Figure 23 illustrates the relationship pattern. The following patterns can be examined visually:

- **Repeated involvement:** one accused connected to several cases.
- **Connections between cases:** two cases sharing an accused, station or crime head.
- **Relationships between entities:** the path accused - case - victim.
- **Clusters:** groups of records densely connected around common entities.
- **Potentially significant patterns:** for example concentration of linked cases in one station or district.

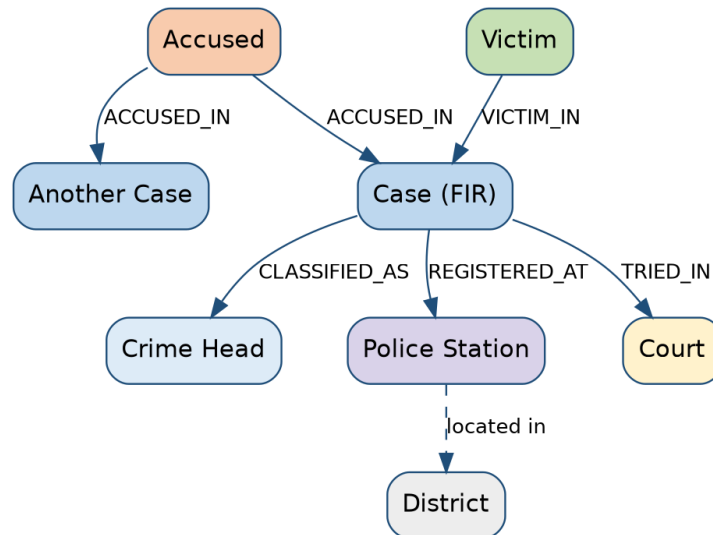


Figure 23: Criminal Network Relationship Diagram (Conceptual)

Network analysis in the literature offers structural measures and role analysis [2], [3]; this system presents the graph but does not compute such measures. A link in the graph records that an association exists in the database. It does not establish that any person committed an offence, and it carries no legal conclusion.

8.10 Limitations of AI-based Analysis

- **Data dependence:** outputs reflect what has been recorded, including any recording bias [10].
- **Small dataset:** 120 FIRs limit statistical stability.
- **Hallucination:** generated text may include unsupported statements [7].
- **No evaluation:** no precision, recall or accuracy has been measured; none is claimed.
- **Opaque scores:** risk scores and confidence labels are not calibrated probabilities.
- **False positives and negatives:** flagged hotspots or anomalies may be spurious, and real patterns may be missed.

8.11 Ethical Considerations

- The AI assists analysts. Its output is decision support and not a decision.
- The AI does not establish guilt; associations in a graph are not evidence.
- Human verification of every AI-derived statement is required before action.

- Bias and false positives must be considered, since predictive policing can reinforce historical patterns [10].
- Personal data must be protected and the use of names in demonstrations should be confined to synthetic or authorised data.
- Operational suggestions such as patrol deployments generated by AI, as shown in the hotspot view, are advisory and should not replace professional judgement [8].

CHAPTER 9 – TESTING

9.1 Testing Introduction

Testing establishes whether the system behaves as specified. This chapter reports only tests whose results are known. One test, the production KPI API call, was executed and recorded. Other planned tests are listed without results, and observations from screenshots are labelled as such and are not counted as pass results.

9.2 Testing Strategy

The strategy combines unit testing of backend functions, integration testing across frontend, API and database, API testing of endpoints, database testing of queries and pooling, frontend testing of rendering, CORS testing across origins, and deployment testing on Vercel. Automated test suites, if any exist in the repository, are [TO BE PROVIDED]. User-interface testing of layout and navigation, and user acceptance testing with representative users, are also planned; no results are available [TO BE PROVIDED].

9.3 Unit Testing

Unit tests would exercise DatabaseHelper methods and groq_service.py functions in isolation (with the Groq call mocked). No unit-test results are available: [TO BE PROVIDED].

9.4 Integration Testing

The production KPI call is an integration test in effect, because a single request traversed the Vercel-hosted API, the database connection and the JSON response path. Other integration tests are [TO BE PROVIDED].

9.5 API Testing

API testing sends HTTP requests to endpoints and checks the status code and body. The KPI endpoint was tested against the production backend and returned HTTP 200 with the JSON body shown in Section 6.9.

9.6 Database Testing

Database testing would check schema, constraints, query correctness and pool behaviour under concurrent requests. The KPI response is consistent with a functioning database connection. Further results are [TO BE PROVIDED].

9.7 Frontend Testing

Screenshots show that the login, registration, dashboard, case list, map, analytics, AI, network, officer and approval pages render with data. This is observational evidence and not a systematic test.

9.8 CORS Testing

CORS testing confirms that requests from the deployed frontend origin succeed and that requests from unauthorised origins are refused. The dashboard displaying live values indicates that cross-origin calls from the frontend succeeded in the tested configuration; a negative test is [TO BE PROVIDED].

9.9 Deployment Testing

The verified production KPI request is evidence that the backend was reachable in production, and the Vercel dashboards show both projects in the Ready state with a 0 percent error rate for the observed window. Additional checks of the Vercel build, environment variables and serverless cold-start behaviour are [TO BE PROVIDED].

9.10 Test Cases

Table 8 lists the recorded test and the planned tests. Only T-01 has a recorded test result with a captured response body. Rows marked "Observed" rely on supplied screenshots; rows marked "Not recorded" were not executed or their results were not supplied.

Table 8: Test Cases

Test ID	Test Description	Input	Expected Result	Actual Result	Status
T-01	Production KPI API (integration)	GET /api/cases/kpis	HTTP 200 + JSON response	HTTP 200 + verified KPI JSON (Section 6.9)	PASS
T-02	Dashboard shows KPI values	Open Command view	Cards show API values	Screenshot shows 120 / 110 / 79 / 29 / 7 / +1 today	Observed

Test ID	Test Description	Input	Expected Result	Actual Result	Status
T-03	Case list with filters	Open FIR Management	Table of cases with filters	Screenshot shows table and filters	Observed
T-04	Network graph for a selected accused	Select A42 - Ramesh Shetty	Isolated subgraph	Screenshot shows victim, accused, case, crime head, court, station nodes	Observed
T-05	Hotspot prediction view	Open AI Intelligence Modules	Ranked hotspots	Screenshot shows ranked items with risk scores	Observed
T-06	Sign-in with valid credentials	Valid user	Access to portal	Screenshots show portal access as administrator	Observed
T-07	Sign-in with invalid credentials	Invalid user	Access denied	[TO BE PROVIDED]	Not recorded
T-08	Unauthorised access to protected endpoint	Request without credentials	HTTP 401/403	[TO BE PROVIDED]	Not recorded
T-09	AI chat request	Question on most common crime categories	Relevant response	Screenshot shows Groq online and a category table (30 / 24 / 23 cases)	Observed
T-10	AI fallback on primary model failure	Simulated failure	Fallback model used	[TO BE PROVIDED]	Not recorded
T-11	CORS from disallowed origin	Cross-origin request	Blocked	[TO BE PROVIDED]	Not recorded
T-12	Concurrent requests with pooling	Parallel requests	No connection exhaustion	[TO BE PROVIDED]	Not recorded
T-13	Accused and victim analytics pages render	Open both pages	Registries and summary cards	Screenshots show 79 profiles, 12 flagged; 120 victims, 23 police personnel	Observed
T-14	Production deployment status	Open Vercel dashboards	Ready deployments	Both projects Ready; backend 162 requests, 0% error rate	Observed
T-15	Security role selection	Switch role from Administrator to Police Officer	Role changes and the interface reflects it	[TO BE PROVIDED]	Not recorded
T-16	Global search by crime number	Valid crime number	Matching case returned	[TO BE PROVIDED]	Not recorded
T-17	Case network isolation	Select a case in Isolate Case Network	Connected network displayed	[TO BE PROVIDED]	Not recorded

CHAPTER 10 – RESULTS AND DISCUSSION

10.1 System Results

The delivered system provides the modules described in Chapter 7. The results reported below are of three kinds: verified API output, database statistics displayed in the dashboard, and observed interface behaviour from screenshots. No AI performance metric is reported because none was measured.

10.2 Production API Results

The production backend endpoint GET /api/cases/kpis was successfully tested and returned HTTP 200. The verified values are given in Table 9. They are presented as verified system/database statistics at the time of testing.

Table 9: Production KPI Results (Verified System/Database Statistics)

Metric	Value
Total FIRs	120
Solved Cases	110
Closed Cases	59
Pending Cases	10
Charge-sheeted Cases	51
Accused	79
Victims	120
Districts	7
Police Stations	29
Today's FIRs	1

These figures are system/database statistics. They are not AI accuracy metrics, model performance metrics, precision, recall or any other evaluation of the AI functionality.

Two internal consistencies can be observed: solved cases (110) equal charge-sheeted (51) plus closed (59) cases, and solved (110) plus pending (10) equal total FIRs (120). The dashboard expresses solved over total as a resolution rate of 92 percent.

10.3 Dashboard Results

Figure 10 shows the values above rendered as cards, with a monthly crime volume chart. In the monthly series the highest peak appears around mid-2025 at approximately nine

cases in a month and typical months lie between one and six cases, as read from the chart. The crime-by-district chart in Figure 16 shows Bengaluru City as the largest of seven districts, and the victim gender chart shows 77 percent male and 23 percent female. These observations describe the recorded data only.

10.4 Database Results

The consistent KPI response and the presence of related records (cases, accused, victims, officers, stations) across many screens demonstrate a working PostgreSQL integration. Query performance and pool behaviour under load were not measured and are [TO BE PROVIDED].

10.5 AI Functionality

The AI chat page (Figure 21) answers a category question with counts drawn from the 120 records. The hotspot tab lists ranked locations with risk scores (for example 79 out of 100 for the top-ranked station) and suggested actions. The network view builds linkage graphs. These demonstrate that the AI-assisted features run and produce outputs. They do not show that the outputs are correct; correctness evaluation against ground truth was not performed.

10.6 Deployment Results

The production KPI response confirms that the backend deployed on Vercel is reachable and connected to its database. The Vercel dashboards (Figure 8 and Figure 9) show the backend and frontend projects in the Ready state on the main branch, with the backend recording 162 function invocations and a 0 percent error rate over 6 hours.

Table 10: Implemented, Verified and Future Functionality

Item	Implemented	Verified	Future
KPI endpoint and dashboard cards	Yes	Yes (T-01)	-
Case list, map, analytics, officer, accused and victim views	Yes (screenshots)	Observed only	-
Hotspot, forecast, anomaly tabs	Yes (screenshot of hotspot tab)	No	Evaluation
Criminal network graph	Yes (screenshots)	Observed only	Graph algorithms
AI chat	Yes (screenshot)	Observed only	Grounding, explainability
Login, request and approval workflow	Yes (screenshots)	Observed only	Fine-grained RBAC, full audit logging

Item	Implemented	Verified	Future
Graph neural networks, streaming, mobile app	No	No	Yes

10.7 Discussion

The system demonstrates a complete path from a relational database to analytical and relationship views on a public deployment. Its strengths are integration and the relationship graph. Its principal weaknesses are the small data volume (120 FIRs), the lack of quantitative AI evaluation, and the incomplete evidence about security enforcement. Interpretations of hotspots and risk scores should be cautious for the reasons discussed in Section 8.10 and [10].

CHAPTER 11 – ADVANTAGES, LIMITATIONS AND APPLICATIONS

11.1 Advantages

Table 11: Advantages

Advantage	Explanation
Centralised data	Cases, accused, victims, officers and stations in one relational database
Relationship visibility	Typed graph shows links between entities
Layered architecture	Separable frontend, API, database and AI service
Accessible	Browser-based; REST API reusable by other clients
Decision support	Ranked hotspots, anomalies and AI summaries help prioritise review
Managed deployment	Serverless hosting reduces operational overhead

11.2 Limitations

Table 12: Limitations

Limitation	Explanation
Data quality dependence	Outputs are only as reliable as the records
AI may be incorrect or incomplete	Generated text and scores can be wrong; [7]
Human verification required	No output should be acted on without review
Data privacy requirements	Records concern identifiable persons
Authentication and authorisation	A production system needs proper authentication, authorisation and auditing on every endpoint
Limited dataset scope	120 FIRs, 7 districts, 29 stations
No AI evaluation	No accuracy or performance metrics were measured
Basic network analysis	No centrality, community detection or graph learning
Incomplete relationships	Missing or incomplete relationships in the source data produce an incomplete network view
Integration and load behaviour	Integration with other departmental systems and behaviour under large data volumes or many concurrent users are [TO BE PROVIDED]

11.3 Applications

- Crime data analysis
- Police intelligence support
- Case relationship analysis

- District-level crime analysis
- Investigation support
- Crime trend monitoring
- Academic demonstration of full-stack AI systems

11.4 Ethical and Legal Considerations

Crime records contain personal data of accused persons, victims and officers. Use in a real setting requires legal authority, purpose limitation, access control, retention rules and auditability. The interface uses the name and branding of the Karnataka State Police; unless authorised, such branding must not imply an official system, and whether its use is authorised is [TO BE PROVIDED]. Demonstration data should be synthetic or anonymised, and its provenance should be stated: the origin of the dataset is [TO BE PROVIDED]. The screenshots show full names of accused persons and victims, and a victim registry with ages; these must be confirmed as synthetic before the report is circulated. Algorithmic outputs should never be used as evidence of guilt; associations require independent investigation, and the consequences of false positives fall on individuals and communities [10].

In addition, the following principles govern the intended use of the system:

- **Presumption of innocence:** no output of the system determines guilt.
- **Human oversight:** every analytical lead requires verification by a qualified investigator.
- **Explainability:** the basis of an AI-derived alert should be traceable to source data wherever possible.
- **Misuse prevention:** network and search capabilities must not be used for purposes outside authorised investigation.
- **Evidence verification:** any lead drawn from the network view must be independently verified before it is treated as evidence.

The AI-Powered Criminal Network Analysis System is designed as an investigation-support platform. Its analytical outputs should be treated as leads or insights requiring appropriate human verification and should not independently determine criminal responsibility or guilt.

CHAPTER 12 – FUTURE ENHANCEMENT

The items below are FUTURE WORK. None is claimed as current functionality.

Table 13: Future Enhancements (Not Implemented)

Enhancement	Description
Graph neural networks	Learning on the entity graph, for example graph convolutional networks [24]
Advanced graph algorithms	Centrality, community detection and link prediction [4]
Real-time data integration	Live ingestion from source systems
Improved predictive models	Statistically validated space-time models [12]
Streaming analytics	Event-stream processing for continuous updates
Advanced anomaly detection	Evaluated detectors for point, contextual and collective anomalies [13]
Role-based access control	Server-enforced permissions per role and district
Audit logging	Immutable record of access and actions
Mobile application	Field access for authorised users
Multilingual support	Kannada and other languages in interface and chat
Improved visualisation	Larger graphs, filtering and time-sliced views
Explainable AI	Show the evidence behind each score and grounded answers [9]
Secure government/police integration	Approved interfaces, hardened hosting and compliance review
Formal evaluation	Benchmarks with ground truth for hotspot, anomaly and similarity outputs
Temporal network analysis	Tracking how relationships between entities evolve over time
Advanced geospatial analytics	Further spatial analysis layered onto the Interactive Map module
Voice interface	Voice-based interaction with the AI assistant

CHAPTER 13 – CONCLUSION

Problem. Crime records contain relationships that are hard to see in record-level views, and analysts need integrated trend, spatial and relationship views.

Solution and architecture. The AI-Powered Criminal Network Analysis System provides a browser-based platform built from a React and TypeScript frontend, a Flask REST backend and a PostgreSQL database, with the frontend reaching the backend through a centralised API service configured by VITE_API_URL.

AI integration. A Groq-based service (default model openai/gpt-oss-120b, with configured fallbacks) supplies AI-assisted analysis and chat. It is positioned as decision support, and human verification is required.

Database and API. PostgreSQL with connection pooling stores the records; a REST API exposes cases, KPIs, trends, geography, entities, AI analytics and the criminal network.

Deployment and verified results. The system is deployed on Vercel. The production endpoint GET /api/cases/kpis returned HTTP 200 with 120 FIRs, 110 solved, 59 closed, 10 pending, 51 charge-sheeted, 79 accused, 120 victims, 7 districts, 29 police stations and 1 FIR for the day. These are database statistics, not AI accuracy.

Contribution. The project contributes an integrated, deployed prototype that connects relational crime data to analytics, maps, AI assistance and a relationship graph. Its limitations, notably the small dataset, the lack of quantitative AI evaluation and the need for verified authorisation, define clear directions for further work.

REFERENCES

- [1] H. Chen, W. Chung, J. J. Xu, G. Wang, Y. Qin, and M. Chau, "Crime data mining: A general framework and some examples," *Computer*, vol. 37, no. 4, pp. 50-56, Apr. 2004.
- [2] J. Xu and H. Chen, "Criminal network analysis and visualization," *Commun. ACM*, vol. 48, no. 6, pp. 100-107, Jun. 2005.
- [3] M. K. Sparrow, "The application of network analysis to criminal intelligence: An assessment of the prospects," *Social Networks*, vol. 13, no. 3, pp. 251-274, 1991.
- [4] M. E. J. Newman, *Networks: An Introduction*. Oxford, U.K.: Oxford Univ. Press, 2010.
- [5] S. Wasserman and K. Faust, *Social Network Analysis: Methods and Applications*. Cambridge, U.K.: Cambridge Univ. Press, 1994.
- [6] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017, pp. 5998-6008.
- [7] Z. Ji et al., "Survey of hallucination in natural language generation," *ACM Comput. Surv.*, vol. 55, no. 12, Art. no. 248, 2023.
- [8] W. L. Perry, B. McInnis, C. C. Price, S. C. Smith, and J. S. Hollywood, *Predictive Policing: The Role of Crime Forecasting in Law Enforcement Operations*. Santa Monica, CA, USA: RAND Corporation, 2013.
- [9] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020, pp. 9459-9474.
- [10] K. Lum and W. Isaac, "To predict and serve?," *Significance*, vol. 13, no. 5, pp. 14-19, Oct. 2016.
- [11] L. W. Sherman, P. R. Gartin, and M. E. Buerger, "Hot spots of predatory crime: Routine activities and the criminology of place," *Criminology*, vol. 27, no. 1, pp. 27-56, 1989.
- [12] G. O. Mohler, M. B. Short, P. J. Brantingham, F. P. Schoenberg, and G. E. Tita, "Self-exciting point process modeling of crime," *J. Amer. Statist. Assoc.*, vol. 106, no. 493, pp. 100-108, 2011.
- [13] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Comput. Surv.*, vol. 41, no. 3, Art. no. 15, Jul. 2009.
- [14] PostgreSQL Global Development Group, "PostgreSQL Documentation." [Online]. Available: <https://www.postgresql.org/docs/> (accessed Sep. 2026).
- [15] R. T. Fielding, "Architectural styles and the design of network-based software architectures," Ph.D. dissertation, Univ. California, Irvine, CA, USA, 2000.
- [16] Meta Open Source, "React Documentation." [Online]. Available: <https://react.dev/> (accessed Sep. 2026).
- [17] Vite Team, "Vite Documentation." [Online]. Available: <https://vite.dev/> (accessed Sep. 2026).

- [18] Pallets Projects, "Flask Documentation." [Online]. Available: <https://flask.palletsprojects.com/> (accessed Sep. 2026).
- [19] C. Hadd et al., "Flask-CORS Documentation." [Online]. Available: <https://flask-cors.readthedocs.io/> (accessed Sep. 2026).
- [20] B. Chesneau, "Gunicorn Documentation." [Online]. Available: <https://docs.gunicorn.org/> (accessed Sep. 2026).
- [21] The Psycopg Team, "Psycopg 2 Documentation." [Online]. Available: <https://www.psycopg.org/docs/> (accessed Sep. 2026).
- [22] Groq, Inc., "GroqCloud Documentation." [Online]. Available: <https://console.groq.com/docs> (accessed Sep. 2026).
- [23] Vercel, Inc., "Vercel Documentation." [Online]. Available: <https://vercel.com/docs> (accessed Sep. 2026).
- [24] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in Proc. Int. Conf. Learn. Represent. (ICLR), 2017.